

MAXIMAL OPERATORS AND
THE FEFFERMAN-STEIN INEQUALITY
ON BOUNDED DOMAINS

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THESIS: MAXIMAL OPERATORS AND
THE FEFFERMAN-STEIN INEQUALITY
ON BOUNDED DOMAINS

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ABSTRACT

This thesis studies the Hardy-Littlewood maximal function, the restricted and sharp maximal function, and the local Fefferman–Stein inequality on a class of irregular domains. We begin by developing the theory of maximal functions and their applications in Sobolev spaces, including regularity, convolution approximations, and the identification of Lipschitz functions. We then introduce the restricted sharp maximal function and investigate its role in establishing local Fefferman–Stein inequalities. We also show that the classical unweighted inequality fails on the cuspidal domain by demonstrating its implication of the improved Poincaré inequality, which does not hold in non-John domains. We then prove a weaker weighted version of the local Fefferman–Stein inequality on the cuspidal domain. Our approach relies on a decomposition of the domain, followed by localized analysis on each component and a local-to-global argument to piece together the global estimate.

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Introduction

The theory of maximal operators and their applications lies at the heart of many modern developments in real analysis and partial differential equations. This thesis is concerned with the study of the Hardy–Littlewood maximal function, its role in Sobolev space theory, and the extension of related inequalities to irregular domains.

In Chapter 1, we introduce the maximal operator, present its fundamental properties, and discuss the Hardy–Littlewood maximal theorem concerning continuity in $L^p(\mathbb{R}^n)$. The chapter also explores an important application: the proof of the Lebesgue Differentiation Theorem, highlighting the pivotal role the maximal function plays in connecting measure theoretic properties with pointwise behavior. Additionally, we define the restricted sharp maximal operator, a concept that will play a central role in subsequent chapters.

Chapter 2 focuses on the application of maximal functions in the setting of first-order Sobolev spaces. Sobolev spaces are essential for analyzing functions that may lack classical differentiability but possess generalized, or weak, derivatives. We introduce the notions of weak partial derivatives and Sobolev spaces and examine how maximal functions assist in characterizing Lipschitz functions and approximating Sobolev functions by smoother ones. We also introduce the space of functions of bounded mean oscillation (BMO) and explore how certain Sobolev functions can be identified as elements of BMO through the restricted sharp maximal operator.

In Chapter 3, we turn our attention to the Fefferman–Stein inequality, particularly in non1

regular domains. After reviewing the classical Fefferman–Stein inequality on John domains, we consider a planar domain that does not satisfy the John condition, making the classical approach unknown. We develop a weaker, weighted version of the local Fefferman–Stein inequality for this cuspidal domain, utilizing decomposition techniques and a local to global argument.

Throughout this thesis, we aim to highlight the interplay between maximal functions, Sobolev spaces, and domain geometry, providing new insights into how analytic inequalities can adapt to irregular settings.

Chapter 1

Maximal operators

This chapter is devoted to the maximal operator and its applications. We define the Maximal operator in Section 1.1 and introduce properties along with the Hardy-Littlewood Maximal function theorem, which deals with the continuity of this operator in $L^p(\mathbb{R}^n)$. In Section 1.2, we study an application of the maximal operator in proving the Lebesgue Differentiation Theorem. In Section 1.3, we introduce the restricted sharp maximal operator, which will be an important concept in this thesis. The results presented here are based on [5, Chapter 1].

1.1 The Centered Hardy-Littlewood maximal operator

We recall the function spaces $L^p(\Omega)$, $L_0^p(\Omega)$ and $L_{\text{loc}}^p(\Omega)$.

Definition 1.1.1. Let $\Omega \subset \mathbb{R}^n$ be a measurable set and $1 \leq p \leq \infty$. The space of all measurable functions $f : \Omega \rightarrow \mathbb{R}$ such that

$$\left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}} < \infty$$

(with the usual modification when $p = \infty$) is denoted by $L^p(\Omega)$ and is pronounced *el-p*. This

space is equipped with the norm

$$\|f\|_{L^p(\Omega)} := \left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}}.$$

The subspace $L_0^p(\Omega) \subset L^p(\Omega)$ consists of functions with vanishing mean value over Ω , that is,

$$\int_{\Omega} f(x) dx = 0.$$

Finally, the space $L_{\text{loc}}^p(\Omega)$ consists of functions f such that for every compact set $K \subset \Omega$, the local L^p -norm is finite:

$$\left(\int_K |f(x)|^p dx \right)^{\frac{1}{p}} < \infty.$$

We now define an important function that will be used throughout this thesis and plays a central role in harmonic analysis. This function is known as the *Hardy–Littlewood maximal function*, or simply the *maximal function*, depending on the context.

Definition 1.1.2. Suppose a function $f \in L_{\text{loc}}^1(\mathbb{R}^n)$. The centered Hardy-Littlewood maximal function $Mf : \mathbb{R}^n \rightarrow [0, \infty]$ is defined by

$$Mf(x) = \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| dy$$

for every $x \in \mathbb{R}^n$.

We may think of the Hardy-Littlewood maximal function as the supremum of the averages of the absolute value of a function on open balls centered at a point. In chapters 2 and 3 of this thesis, we will use cubes instead of open balls, specifically half-open cubes. Therefore, we introduce a non-centered Hardy-Littlewood maximal function on cubes.

Definition 1.1.3. Suppose $f \in L_{\text{loc}}^1(\mathbb{R}^n)$. The non-centered Hardy-Littlewood Maximal function on cubes $M^*f : \mathbb{R}^n \rightarrow [0, \infty]$ is defined by

$$M^*f(x) = \sup_{Q \ni x} \int_Q |f(y)| dy$$

for every $x \in \mathbb{R}^n$.

We are able to compare both versions of the Hardy-Littlewood maximal function through the estimate

$$\frac{1}{C}Mf(x) \leq M^*f(x) \leq CMf(x).$$

where $C > 0$ and $C = C(n)$. For more information on this inequality, see [[5], pg. 2].

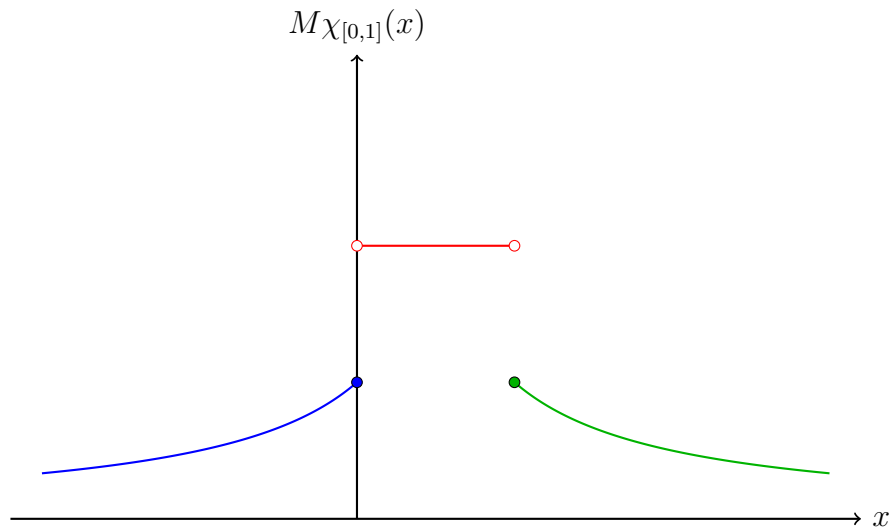
Example 1.1.4. To exemplify the above definition, consider as $f(x)$ the characteristic function $\chi_{[0,1]}(x)$ of the interval $[0, 1]$,

$$\chi_{[0,1]}(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

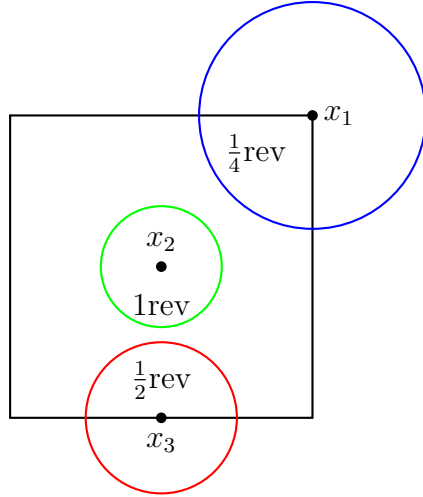
It can be verified the maximal function in this case is the following:

$$M\chi_{[0,1]}(x) = \begin{cases} \frac{1}{2(1-x)} & \text{if } x \leq 0, \\ 1 & \text{if } 0 < x < 1, \\ \frac{1}{2x} & \text{if } x \geq 1. \end{cases}$$

We have included below the graph of the maximal function where it is possible to observe the discontinuities that appear on the boundary of the interval $[0, 1]$.



Example 1.1.5. The example shown above appears in some references about the maximal operator. Notice that it is discontinuous at the boundary of the domain $[0, 1] \subset \mathbb{R}$. In the following example, which we have not seen in literature, we study the maximal function of the characteristic of a convex domain in $\mathbb{R}^2$. Note that the maximal function is also discontinuous on the boundary and we exhibit a relation between the values on ∂U and the section angle. Let $U \subseteq \mathbb{R}^2$ be a convex polygon in the plane and $f(x) = \chi_U(x)$ its characteristic function. We conjecture that $Mf(x)$ is equal to the interior angle of ∂U at the point x measured in revolutions. For example, if U is a square in the plane then the maximal of the characteristic function of U takes the value $1/4$ on the four corners of the square and $1/2$ in its four open sides, and 1 in the interior.



It is appropriate to include some properties for the centered maximal as an operator. These results show that Mf is a sublinear operator.

Proposition 1.1.6. Assume that $f, g \in L^1_{\text{loc}}(\mathbb{R}^n)$ and let $x \in \mathbb{R}^n$. Then the following hold:

- (a) $Mf(x) \geq 0$,
- (b) $M(f + g)(x) \leq Mf(x) + Mg(x)$,
- (c) $|Mf(x) - Mg(x)| \leq M(f - g)(x)$ if $Mf(x) < \infty$ and $Mg(x) < \infty$,

- (d) $M(af)(x) = |a|Mf(x)$, for every $a \in \mathbb{R}$,
- (e) $M(\tau_h f)(x) = (\tau_h Mf)(x)$, where $\tau_h f(x) = f(x + h)$ with $h \in \mathbb{R}^n$,
- (f) $M(\delta_a f)(x) = (\delta_a Mf)(x)$, where $\delta_a f(x) = f(ax)$ with $a > 0$.

Proof. Assertion (a) follows from the definition of the Maximal operator. To prove assertion (b), we note that

$$\begin{aligned}
M(f + g)(x) &= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) + g(y)| \, dy \\
&\leq \sup_{r>0} \left(\frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| \, dy + \frac{1}{|B(x, r)|} \int_{B(x, r)} |g(y)| \, dy \right) \\
&\leq \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| \, dy + \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |g(y)| \, dy \\
&= Mf(x) + Mg(x).
\end{aligned}$$

To prove (c), we assume $Mf(x) < \infty$ and $Mg(x) < \infty$. By property (b), we have that

$$Mf(x) = M(f - g + g)(x) \leq M(f - g)(x) + Mg(x).$$

Therefore,

$$|Mf(x) - Mg(x)| \leq M(f - g)(x).$$

To prove assertion (d), we have that

$$\begin{aligned}
M(af)(x) &= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |af(y)| \, dy \\
&= |a| \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| \, dy = |a|Mf(x)
\end{aligned}$$

for every $a \in \mathbb{R}$, with the convention that $0 \cdot \infty = 0$. Assertion (e) follows from a change of variables, where $y = z - h$,

$$\begin{aligned}
M(\tau_h f)(x) &= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y + h)| \, dy \\
&= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x+h, r)} |f(z)| \, dz \\
&= \sup_{r>0} \frac{1}{|B(x + h, r)|} \int_{B(x+h, r)} |f(z)| \, dz = \tau_h Mf(x).
\end{aligned}$$

Finally, to prove assertion (f), we use another change of variables with $y = \frac{z}{a}$,

$$\begin{aligned} M(\delta_a)(x) &= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(ay)| \, dy \\ &= \sup_{r>0} \frac{1}{|B(x, r)|} \int_{B(ax, ar)} |f(z)| a^{-n} \, dz \\ &= \sup_{r>0} \frac{1}{|B(ax, ar)|} \int_{B(ax, ar)} |f(z)| \, dz = \delta_a Mf(x). \end{aligned}$$

□

A result of Lemma 1.1.6 shows that Mf is a continuous function, given that Mf is finite and f is continuous.

Corollary 1.1.7. Assume that f is a continuous function in $\mathbb{R}^n$. If $Mf(x) < \infty$ for every $x \in \mathbb{R}^n$, then Mf is continuous in $\mathbb{R}^n$.

Proof. Let $x, h \in \mathbb{R}^n$. Since $Mf(x) < \infty$ and by Lemma 1.1.6 properties (c), (b), and (e), we have

$$|M(\tau_h f)(x) - Mf(x)| \leq M(\tau_h f - f)(x) \leq Mf(x + h) + Mf(x) < \infty.$$

Let $\epsilon > 0$. Then there exists a ball $B(x, r)$ such that

$$M(\tau_h f - f)(x) \leq \frac{1}{|B(x, r)|} \int_{B(x, r)} |\tau_h f(y) - f(y)| \, dy + \frac{\epsilon}{2}.$$

Since f is uniformly continuous in $\overline{B}(x, r + 1)$, there is a $0 < \delta < 1$ such that

$$|\tau_h f(y) - f(y)| = |f(y + h) - f(y)| < \frac{\epsilon}{2}$$

whenever $y \in B(x, r)$ and $|h| < \delta$. Therefore,

$$\begin{aligned} |M(\tau_h f)(x) - Mf(x)| &\leq M(\tau_h f - f)(x) \\ &\leq \frac{1}{|B(x, r)|} \int_{B(x, r)} |\tau_h f(y) - f(y)| \, dy + \frac{\epsilon}{2} \\ &< \epsilon \end{aligned}$$

whenever $|h| < \delta$. By Lemma 1.1.6, part (e),

$$|Mf(x+h) - Mf(x)| = |(\tau_h Mf)(x) - Mf(x)| = |M(\tau_h f)(x) - Mf(x)| < \epsilon$$

whenever $|h| < \delta$. This implies that Mf is continuous. $\square$

A fundamental result in Harmonic analysis is the Hardy-Littlewood-Wiener maximal function theorem which states that the centered Hardy-Littlewood maximal function is continuous in $L^p(\mathbb{R}^n)$ with $1 < p \leq \infty$. For $p = 1$ we obtain a weak-type estimate that controls the size of the set where the maximal function is larger than a positive real number. For the proof we require the Vitali $5r$ covering lemma.

Lemma 1.1.8. Let $\mathcal{B}$ be a collection of open balls $B(x, r_x) \in \mathbb{R}^n$, with radius r_x . Assume that the balls in $\mathcal{B}$ have uniformly bounded radii. Then there exists a countable subcollection of pairwise disjoint balls $B(x_i, r_i) \in \mathcal{B}$, $i \in \mathbb{N}$, such that

$$\bigcup_{B \in \mathcal{B}} B \subseteq \bigcup_{i=1}^{\infty} B(x_i, 5r_i).$$

For a proof of this lemma, see [10, Pg. 23]. The significance of this lemma lies in its ability to extract a countable sub-collection that covers the union of the original open balls. Although we expand the open balls in the countable subcollection, the Vitali $5r$ covering lemma helps us compare the measure of a union of a collection of open balls with the measure of a countable collection of open balls.

We also require the following lemma, that evaluates the $L^p(A)$ norm of a function with distribution sets.

Lemma 1.1.9. Let $0 < p < \infty$. Assume that $A \subset \mathbb{R}^n$ is a measurable set and $f : A \rightarrow [-\infty, \infty]$ is a measurable function. Then

$$\int_A |f(x)|^p = p \int_0^\infty t^{p-1} |\{x \in A : |f(x)| > t\}| dt.$$

Proof. By Fubini's theorem,

$$\begin{aligned}
\int_A |f(x)|^p dx &= p \int_{\mathbb{R}^n} \chi_A(x) \int_0^{|f(x)|} t^{p-1} dt dx \\
&= p \int_{\mathbb{R}^n} \int_0^\infty \chi_A(x) \chi_{[0, |f(x)|)}(t) t^{p-1} dt dx \\
&= p \int_0^\infty \int_{\mathbb{R}^n} \chi_A(x) \chi_{[0, |f(x)|)}(t) t^{p-1} dx dt \\
&= p \int_0^\infty t^{p-1} \int_{\mathbb{R}^n} \chi_A(x) \chi_{\{x \in \mathbb{R}^n : |f(x)| > t\}}(x) dx dt \\
&= p \int_0^\infty t^{p-1} \int_{\mathbb{R}^n} \chi_{\{x \in A : |f(x)| > t\}}(x) dx dt \\
&= p \int_0^\infty t^{p-1} |\{x \in A : |f(x)| > t\}| dt.
\end{aligned}$$

□

Both Lemma 1.1.8 and 1.1.9 are applied in the proof for the following Hardy-Littlewood Maximal function Theorem.

Theorem 1.1.10. Let $1 \leq p \leq \infty$ and assume that $f \in L^p(\mathbb{R}^n)$. The the following holds:

- (a) Mf is finite almost everywhere.
- (b) If $p = 1$, then

$$|\{x \in \mathbb{R}^n : Mf(x) > t\}| \leq \frac{5^n}{t} \int_{\mathbb{R}^n} |f(x)| dx$$

for all $t > 0$.

- (c) If $1 < p \leq \infty$, then $Mf \in L^p(\mathbb{R}^n)$ and

$$\|Mf\|_{L^p(\mathbb{R}^n)} \leq 2 \cdot \left(\frac{5^n p}{p-1} \right)^{\frac{1}{p}} \|f\|_{L^p(\mathbb{R}^n)}.$$

Proof. Let us prove assertion (b). We assume that $f \in L^1(\mathbb{R}^n)$ and consider the set

$$E_t = \{x \in \mathbb{R}^n : Mf(x) > t\}$$

for $t > 0$. From the definition of Mf we have that for every $x \in E_t$ there exists an $r_x > 0$ such that

$$\frac{1}{|B(x, r_x)|} \int_{B(x, r_x)} |f(y)| dy > t.$$

We have that $|B(x, r_x)| = \alpha_n r_x^n$ for some constant α_n . We obtain

$$\int_{B(x, r_x)} |f(y)| dy > t |B(x, r_x)| = t \alpha_n r_x^n. \quad (1.1)$$

That implies

$$r_x^n \leq \frac{1}{t \alpha_n} \int_{B(x, r_x)} |f(y)| dy \leq \frac{1}{t \alpha_n} \int_{\mathbb{R}^n} |f(y)| dy < \infty.$$

We now may construct a collection of open balls $\mathcal{B} = \{B(x, r_x) : x \in E_t\}$ that all have uniformly bounded radii. We also have that $E_t \subseteq \bigcup_{B \in \mathcal{B}} B$. We may now use Lemma 1.1.8 to obtain pairwise disjoint balls $B(x_i, r_i) \in \mathbb{R}^n$ where $i \in \mathbb{N}$ such that

$$E_t \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq \bigcup_{i=1}^{\infty} B(x_i, 5r_i).$$

Therefore,

$$|E_t| \leq \sum_{i=1}^{\infty} |B(x_i, 5r_i)| = 5^n \sum_{i=1}^{\infty} |B(x_i, r_i)|.$$

From 1.1 we have that

$$|E_t| \leq 5^n \sum_{i=1}^{\infty} |B(x_i, r_i)| \leq \frac{5^n}{t} \sum_{i=1}^{\infty} \int_{B(x_i, r_i)} |f(y)| dy \leq \frac{5^n}{t} \int_{\mathbb{R}^n} |f(y)| dy.$$

Hence,

$$|E_t| \leq \frac{5^n}{t} \int_{\mathbb{R}^n} |f(y)| dy.$$

We have proved assertion (b). As $t \rightarrow \infty$, we have that $|\{x \in \mathbb{R}^n : Mf(x) = \infty\}| = 0$, therefore, assertion (a) holds for $p = 1$.

Let us prove assertion (c). Suppose $1 < p < \infty$ and $f \in L^p(\mathbb{R}^n)$ with $t > 0$. We define a function

$$\hat{f}(y) = \begin{cases} f(y) & \text{if } |f(y)| \geq \frac{t}{2}, \\ 0 & \text{otherwise.} \end{cases}$$

We have that $|f| \leq |\hat{f}| + \frac{t}{2}$. Since

$$\begin{aligned} Mf(x) &= \sup_{r>0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y)| dy \\ &\leq \sup_{r>0} \frac{1}{|B(x,r)|} \int_{B(x,r)} \left| \hat{f}(y) + \frac{t}{2} \right| dy = M(\hat{f} + \frac{t}{2})(x), \end{aligned}$$

by Lemma 1.1.6 part (b) we have that $Mf(x) \leq M\hat{f}(x) + \frac{t}{2}$ for every $x \in \mathbb{R}^n$. Consequently,

$$\{x \in \mathbb{R}^n : Mf(x) > t\} \subset \left\{x \in \mathbb{R}^n : M\hat{f}(x) > \frac{t}{2}\right\}.$$

Note that,

$$|\hat{f}(x)| = \chi_{\{y \in \mathbb{R}^n : \hat{f}(y) \neq 0\}}(x) |\hat{f}(x)|^{1-p} |\hat{f}(x)|^p \leq \left(\frac{t}{2}\right)^{1-p} |f(x)|^p,$$

which implies $\hat{f} \in L^1(\mathbb{R}^n)$. By assertion (b),

$$\begin{aligned} |\{x \in \mathbb{R}^n : Mf(x) > t\}| &\leq \left| \left\{x \in \mathbb{R}^n : M\hat{f}(x) > \frac{t}{2}\right\} \right| \\ &\leq \frac{2 \cdot 5^n}{t} \|\hat{f}\|_{L^1(\mathbb{R}^n)} \\ &= \frac{2 \cdot 5^n}{t} \int_{\{x \in \mathbb{R}^n : |f(x)| \geq \frac{t}{2}\}} |f(x)| dx. \end{aligned}$$

By Lemma 1.1.9 and Fubini's Theorem, we have

$$\begin{aligned} \int_{\mathbb{R}^n} (Mf(x))^p dx &= p \int_0^\infty t^{p-1} |\{x \in \mathbb{R}^n : Mf(x) > t\}| dt \\ &\leq p \cdot 2 \cdot 5^n \int_0^\infty t^{p-2} \left(\int_{\{x \in \mathbb{R}^n : |f(x)| \geq \frac{t}{2}\}} |f(x)| dx \right) dt \\ &= p \cdot 2 \cdot 5^n \int_0^\infty t^{p-2} \left(\int_{\mathbb{R}^n} \chi_{\{x \in \mathbb{R}^n : |f(x)| \geq \frac{t}{2}\}}(x) |f(x)| dx \right) dt \\ &= p \cdot 2 \cdot 5^n \int_0^\infty \int_{\mathbb{R}^n} \chi_{[0, 2|f(x)|)}(t) |f(x)| t^{p-2} dx dt \\ &= p \cdot 2 \cdot 5^n \int_{\mathbb{R}^n} \int_0^\infty \chi_{[0, 2|f(x)|)}(t) |f(x)| t^{p-2} dt dx \\ &= p \cdot 2 \cdot 5^n \int_{\mathbb{R}^n} |f(x)| dx \int_0^{2|f(x)|} t^{p-2} dt dx \\ &= p \cdot 2 \cdot 5^n \cdot \frac{2^{p-1}}{p-1} \int_{\mathbb{R}^n} |f(x)| |f(x)|^{p-1} dx. \end{aligned}$$

This proves assertion (c) with $1 < p < \infty$. □

1.2 The Lebesgue Differentiation Theorem

An application of the the Hardy-Littlewood-Wiener maximal function theorem discussed in the previous section allows us to prove the Lebesgue Differentiation Theorem. The proof is interesting because we use the density of continuous functions in $L^p(\mathbb{R}^n)$ to admit convergence for integral averages. This is done through a weak type estimate argument. This theorem tells us that even though we might have a discontinuous or rough function, it's behavior is nice on small sets.

Theorem 1.2.1. Let $1 \leq p < \infty$ and assume that $f \in L^p_{\text{loc}}(\mathbb{R}^n)$. Then

$$\lim_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)|^p dy = 0$$

for almost every $x \in \mathbb{R}^n$.

Proof. If we consider functions $f\chi_{B(0, k)}$, with $k \in \mathbb{N}$, we may assume that $f \in L^p(\mathbb{R}^n)$. Let

$$E_t = \left\{ x \in \mathbb{R}^n : \limsup_{r \rightarrow \infty} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)|^p dy > 2 \cdot 3^p t \right\}$$

for $t > 0$. Since

$$\left\{ x \in \mathbb{R}^n : \limsup_{r \rightarrow \infty} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)|^p dy > 0 \right\} = \bigcup_{j=1}^{\infty} E_{\frac{1}{j}}$$

and

$$\left| \bigcup_{j=1}^{\infty} E_{\frac{1}{j}} \right| \leq \sum_{j=1}^{\infty} |E_{\frac{1}{j}}|,$$

it suffices to show that $|E_t| = 0$ for every $t > 0$.

Let $t > 0$, $x \in E_t$ and $\epsilon > 0$, and let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function satisfying $\|f - g\|_{L^p(\mathbb{R}^n)} < \epsilon$. Since g is continuous, we have

$$\limsup_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |g(y) - g(x)|^p dy = 0,$$

and thus

$$\begin{aligned}
2 \cdot 3^p t &< \limsup_{r \rightarrow \infty} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f(x)|^p dy \\
&\leq 3^p \limsup_{r \rightarrow 0} \left(\frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - g(y)|^p dy + \frac{1}{|B(x, r)|} \int_{B(x, r)} |g(y) - g(x)|^p dy \right) \\
&\quad + 3^p |g(x) - f(x)|^p \\
&= 3^p \limsup_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - g(y)|^p dy + 3^p |f(x) - g(x)|^p \\
&\leq 3^p M(|f - g|^p)(x) + 3^p |f(x) - g(x)|^p.
\end{aligned}$$

Therefore, if $x \in E_t$, then

$$M(|f - g|^p)(x) > t \quad \text{or} \quad |f(x) - g(x)|^p > t,$$

and $E_t \subseteq \{x \in \mathbb{R}^n : M(|f - g|^p)(x) > t\} \cup \{x \in \mathbb{R}^n : |f(x) - g(x)|^p > t\}$. By Theorem 1.1.10 part (b) and Markov's inequality, we obtain

$$\begin{aligned}
|E_t| &\leq |\{x \in \mathbb{R}^n : M(|f - g|^p)(x) > t\}| + |\{x \in \mathbb{R}^n : |f(x) - g(x)|^p > t\}| \\
&\leq \frac{5^n}{t} \int_{\mathbb{R}^n} |f(x) - g(x)|^p dx + \frac{1}{t} \int_{\mathbb{R}^n} |f(x) - g(x)|^p dx \\
&= \frac{5^p + 1}{t} \|f - g\|_{L^p(\mathbb{R}^n)}^p \leq \frac{5^p + 1}{t} \cdot \epsilon^p.
\end{aligned}$$

Letting $\epsilon \rightarrow 0$, we show that $|E_t| = 0$. □

The following definition and theorem will help us prove later results in section 2.2.

Definition 1.2.2. We say that $x \in \mathbb{R}^n$ is a Lebesgue point of $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ if the limit

$$f^*(x) = \lim_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy$$

exists as a real number and

$$\lim_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y) - f^*(x)| dy = 0.$$

The set of Lebesgue points of f is denoted by $\text{Leb}(f)$. Lebesgue points are average values of a function in small neighborhoods of x , and functions do not hyper-oscillate at these points. Note that f need not be defined at x and if $f = g$ almost everywhere, then $f^* = g^*$ almost everywhere. We exemplify a function that has a point in its domain which fails to be a Lebesgue point.

Example 1.2.3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and define

$$H(x) = \begin{cases} 1, & \text{if } x > 0, \\ \frac{1}{2}, & \text{if } x = 0, \\ 0, & \text{if } x < 0. \end{cases}$$

We can verify that

$$\lim_{r \rightarrow 0} \frac{1}{2r} \int_{x-r}^{x+r} f(y) dy = f(x)$$

for every $x \in \mathbb{R}$. However, at $x = 0$, we have that

$$\frac{1}{2r} \int_{-r}^r |f(y) - a| dy = \frac{1}{2r} \int_{-r}^0 |a| dy + \frac{1}{2r} \int_0^r |1 - a| dy = \frac{1}{2}|a| + \frac{1}{2}|1 - a| \neq 0.$$

Therefore, $x = 0$ is not a Lebesgue point.

It turns out that locally integrable functions have Lebesgue points almost everywhere and there exists a subset of $\text{Leb}(f)$ where values of functions equal their Lebesgue points.

Theorem 1.2.4. Assume that $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. Then $|\mathbb{R}^n \setminus \text{Leb}(f)| = 0$. Moreover, there exists a set $E \subset \text{Leb}(f)$ such that $|\mathbb{R}^n \setminus E| = 0$ and $f^*(x) = f(x)$ for every $x \in E$.

An immediate corollary from the previous theorem gives a pointwise estimate for the absolute value of a function by the maximal function

Corollary 1.2.5. Assume that $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. Then $|f(x)| \leq Mf(x)$ for almost every $x \in \mathbb{R}^n$.

Proof. Theorem 1.2.4 implies

$$|f(x)| = |f^*(x)| = \lim_{r \rightarrow 0} \left| \frac{1}{|B(x, r)|} \int_{B(x, r)} f(y) dy \right| \leq \sup_{r > 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} |f(y)| dy = Mf(x).$$

for almost every $x \in \mathbb{R}^n$ □

1.3 The Restricted Maximal Function and the Restricted Sharp Maximal Function

We now introduce two different versions of the Hardy-Littlewood Maximal function that are important in the thesis. Note that these maximal functions are taken over half-open cubes instead of open balls. Cubes add more simplicity in application, for example, in contrast to open balls, cubes can tile a space and have no gaps or overlaps. We will use notation f_Q for the average of a function over a cube Q :

$$f_Q = \frac{1}{|Q|} \int_Q f(x) dx.$$

Definition 1.3.1. Let $\Omega \subset \mathbb{R}^n$. For $f \in L^1(\Omega)$ and $\sigma \geq 1$, we define the restricted maximal function $M_{res, \Omega, \sigma} f : \mathbb{R}^n \rightarrow [0, \infty]$ by

$$M_{res, \Omega, \sigma} f(x) = \sup_{Q \ni x: \sigma Q \subset \Omega} \frac{1}{|Q|} \int_Q |f(y)| dy$$

for all $x \in \Omega$, where we take our supremum over all cubes $Q \subset \mathbb{R}^n$ with $\sigma Q \subset \Omega$ and $x \in Q$.

The following corollary is analogous to Corollary 1.2.5 for a local pointwise estimate on a cube, where the Hardy-Littlewood maximal function is replaced by the restricted version.

Corollary 1.3.2. Let $Q_0 \subset \mathbb{R}^n$ be a cube, and let $f \in L^1_{loc}(Q_0)$. Then for almost every $x \in Q_0$,

$$|f(x)| \leq M_{res, Q_0, 1} f(x).$$

Proof. This follows directly from the Lebesgue differentiation theorem, which holds when averages are taken over cubes. For almost every $x \in Q_0$,

$$|f(x)| = \lim_{\substack{Q \ni x \\ \text{diam}(Q) \rightarrow 0}} \left| \frac{1}{|Q|} \int_Q f(y) dy \right| \leq \sup_{Q \ni x} \frac{1}{|Q|} \int_Q |f(y)| dy = M_{\text{res}, Q_0, 1} f(x)$$

where $\text{diam}(Q) = \sup\{|a - b| : a, b \in Q\}$. □

The restricted maximal function limits the Hardy–Littlewood maximal operator to a set $\Omega \subset \mathbb{R}^n$. However, in many applications—particularly in harmonic analysis—we are also interested in measuring the average deviation of a function from its mean value over subsets of Ω . To capture this kind of local oscillation, the following maximal operator proves to be particularly useful.

Definition 1.3.3. Let $\Omega \subset \mathbb{R}^n$. For $f \in L^1(\Omega)$ and $\sigma \geq 1$, we define the restricted sharp maximal function $M_{\text{res}, \Omega, \sigma}^\# f : \mathbb{R}^n \rightarrow [0, \infty]$ by

$$M_{\text{res}, \Omega, \sigma}^\# f(x) = \sup_{Q \ni x : \sigma Q \subset \Omega} \frac{1}{|Q|} \int_Q |f(y) - f_Q| dy$$

for all $x \in \Omega$, where we take our supremum over all cubes $Q \subset \mathbb{R}^n$ with $\sigma Q \subset \Omega$ and $x \in Q$.

Remark 1.3.4. We note that the restricted sharp maximal operator is monotonic with respect to the partial order of inclusion. Indeed, given two domains $U \subseteq V \subset \mathbb{R}^n$, it follows that for any $f \in L^1(V)$ and $\sigma \geq 1$ the inequalities stated below holds

$$M_{\text{res}, U, \sigma}^\# f(x) \leq M_{\text{res}, V, \sigma}^\# f(x),$$

for any $x \in \mathbb{R}^n$.

Chapter 2

Maximal Function applications in Sobolev Spaces

This chapter explores applications of the Hardy–Littlewood maximal function in the context of first-order Sobolev spaces. Sobolev spaces are fundamental in analysis because they enable the study of less regular functions while retaining a form of differential structure—allowing, in a generalized sense, the performance of calculus on nonsmooth functions. We begin by defining weak partial derivatives and first-order Sobolev spaces in Section 2.1. In Section 2.2, we examine the role of maximal functions in results concerning density and the identification of Lipschitz functions within first-order Sobolev spaces. The chapter concludes with the introduction of the space of functions of Bounded Mean Oscillation (BMO) in Section 2.3, including a result on how certain Sobolev functions can be identified as elements of BMO with respect to a restricted sharp maximal function defined in 1.3.3. The results presented in this chapter are based on foundational work found in [5, Chapter 2].

2.1 Weak Partial Derivatives and first-order Sobolev Spaces

Let $\Omega \subseteq \mathbb{R}^n$ be an open set, and $u \in C^1(\Omega)$. Integration by parts gives

$$\int_{\Omega} u(x) \frac{\partial \phi}{\partial x_i}(x) dx = - \int_{\Omega} \frac{\partial u}{\partial x_i}(x) \phi(x) dx \quad (2.1)$$

for every function $\phi \in C_0^\infty$ and $i = 1, \dots, n$. Note that the boundary values vanish since $\phi \in C_0^\infty(\Omega)$. We then analyze 2.1 when u is not continuously differentiable and note that the right side is not valid. The idea of weak partial derivatives is to find a function $v \in L_{\text{loc}}^1(\Omega)$ that satisfies 2.1, where we denote $v = \frac{\partial u}{\partial x_i}$.

Definition 2.1.1. Let $\Omega \subseteq \mathbb{R}^n$ be an open set, and $u \in L_{\text{loc}}^1(\Omega)$. A function $v \in L_{\text{loc}}^1(\Omega)$ is the i th weak partial derivative of u in Ω , for $i = 1, \dots, n$, if

$$\int_{\Omega} u(x) \frac{\partial \phi}{\partial x_i}(x) dx = - \int_{\Omega} v(x) \phi(x) dx \quad (2.2)$$

for every $\phi \in C_0^\infty(\Omega)$. We denote $v_i = \frac{\partial u}{\partial x_i}$. If all weak partial derivatives exist, with $i = 1, \dots, n$, then

$$\nabla u(x) = \left(\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n} \right),$$

for $x = (x_1, \dots, x_n) \in \Omega$, is the weak gradient of u in Ω .

Example 2.1.2. To exemplify weak derivatives, we consider the function $u : (0, 2) \rightarrow \mathbb{R}$, defined by

$$u(x) = \begin{cases} x & \text{if } 0 < x \leq 1, \\ 1 & \text{if } 1 < x < 2. \end{cases}$$

Note that u is not differentiable at 1. We consider a function $\phi \in C_0^\infty(0, 2)$ and note that

$$\begin{aligned}\int_0^2 u(x)\phi'(x)dx &= \int_0^1 x\phi'(x)dx + \int_1^2 \phi'(x)dx \\ &= -\int_0^1 \phi(x)dx + \phi(1) - \phi(1) \\ &= -\int_0^2 v(x)\phi(x)dx\end{aligned}$$

where

$$v(x) = \begin{cases} 1 & \text{if } 0 < x \leq 1, \\ 0 & \text{if } 1 < x < 2. \end{cases}$$

The function v satisfies 2.2, therefore, it is the weak derivative of u .

Functions that have weak derivatives in $L^p(\Omega)$ for an open set Ω belong to an important Banach space known as the Sobolev space. This space allows us to work with less-smooth functions in partial differentiation theory.

Definition 2.1.3. Let $1 \leq p \leq \infty$ and let $\Omega \subset \mathbb{R}^n$ be an open set. The Sobolev space $W^{1,p}(\Omega)$ contains functions $u \in L^p(\Omega)$, with $u : \Omega \rightarrow \mathbb{R}$, whose weak gradient ∇u exists and $|\nabla u| \in L^p(\Omega)$. For $1 \leq p < \infty$, we equip $W^{1,p}(\Omega)$ with the norm

$$\|u\|_{W^{1,p}(\Omega)} = \left(\|u\|_{L^p(\Omega)}^p + \|\nabla u\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}.$$

We equip the space $W^{1,\infty}(\Omega)$ with the norm

$$\|u\|_{W^{1,\infty}(\Omega)} = \|u\|_{L^\infty(\Omega)} + \|\nabla u\|_{L^\infty(\Omega)}.$$

A function u in Ω belongs to the local Sobolev space $W_{\text{loc}}^{1,p}(\Omega)$ if $u \in W^{1,p}(\Omega')$ for every open set Ω compactly contained in Ω . In some cases, we equip the space $W^{1,p}(\Omega)$ with equivalent norms

$$\left(\|u\|_{L^p(\Omega)}^p + \sum_{i=1}^n \left\| \frac{\partial u}{\partial x_i} \right\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}},$$

for $1 \leq p < \infty$ and

$$\max \left\{ \|u\|_{L^\infty(\Omega)}, \left\| \frac{\partial u}{\partial x_1} \right\|_{L^\infty(\Omega)}, \dots, \left\| \frac{\partial u}{\partial x_n} \right\|_{L^\infty(\Omega)} \right\}$$

for $p = \infty$.

2.2 Density of continuous functions in Sobolev Spaces

In this section we will analyze the role of the Hardy-Littlewood maximal function in proving results for the Sobolev Space $W^{1,p}(\mathbb{R}^n)$. More specifically, the maximal function allows for completion of smooth functions in $W^{1,p}(\mathbb{R}^n)$. This density property helps us prove a theorem which is used to identify Lipschitz functions with the Sobolev Space $W^{1,\infty}(\mathbb{R}^n)$.

Definition 2.2.1. Let f and g be measurable functions where $f, g : [-\infty, \infty] \rightarrow \mathbb{R}^n$. The convolution $f * g$ is defined by

$$(f * g)(x) = \int_{\mathbb{R}^n} f(y)g(x - y)dy$$

for every $x \in \mathbb{R}^n$ for which the integral is defined.

If $1 \leq p \leq \infty$, $f \in L^p(\mathbb{R}^n)$, and $g \in L^1(\mathbb{R}^n)$, then $(f * g)(x)$ exists for almost every $x \in \mathbb{R}^n$ and

$$\|f * g\|_{L^p(\mathbb{R}^n)} \leq \|f\|_{L^p(\mathbb{R}^n)} \|g\|_{L^1(\mathbb{R}^n)}. \quad (2.3)$$

This is known as Young's convolution inequality.

One important function is what some literature refers to as the standard mollifier. Let $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$\varphi(x) = \begin{cases} a \exp\left(\frac{1}{|x|^2 - 1}\right), & \text{if } x \in B(0, 1), \\ 0, & \text{if } x \in \mathbb{R}^n \setminus B(0, 1), \end{cases} \quad (2.4)$$

where the constant a is chosen so that $\int_{\mathbb{R}^n} \varphi(x) dx = 1$. For $x \in \mathbb{R}^n$, $\epsilon > 0$ and $f \in L^1_{\text{loc}}(\mathbb{R}^n)$, we define

$$\varphi_\epsilon(x) = \frac{1}{\epsilon^n} \varphi\left(\frac{x}{\epsilon}\right).$$

Therefore, the convolution

$$(\varphi_\epsilon * f)(x) = \int_{\mathbb{R}^n} \varphi_\epsilon(y) f(x - y) dy \quad (2.5)$$

may be considered as a weighted mean value of the function f in the open ball $B(x, \epsilon)$. This convolution also "smooths" out any function. This is exemplified graphically with the function $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = |x|$. A convolution with the standard mollifier produces a smooth approximation for various values of ϵ (see Figure 2.1).

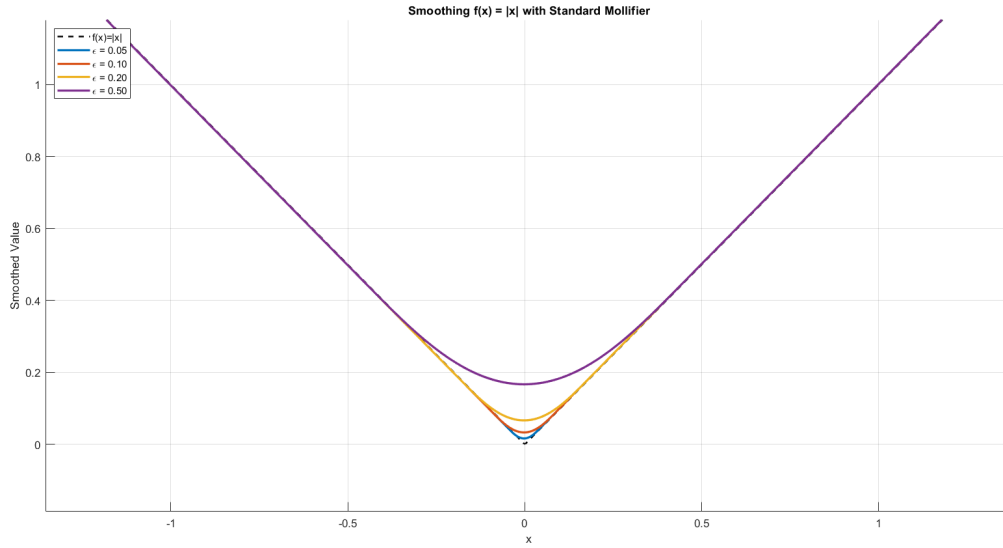


Figure 2.1: The convolution $(\varphi_\epsilon * f)(x)$ where $f(x) = |x|$ on $\mathbb{R}$

Remark 2.2.2. Let $u \in W^{1,p}(\mathbb{R}^n)$ and $\epsilon > 0$. Note that the function φ from 2.4 is in the space $C^\infty_0(\mathbb{R}^n)$, and by the dominated convergence theorem, $\varphi_\epsilon * u \in C^\infty(\mathbb{R}^n) \cap W^{1,p}(\mathbb{R}^n)$

and

$$\frac{\partial(\varphi_\epsilon * u)}{\partial x_i} = \varphi_\epsilon * \frac{\partial u}{\partial x_i} = \frac{\partial \varphi_\epsilon}{\partial x_i} * u$$

for every $i = 1, \dots, n$.

We are able to uniformly control convolution approximations in a pointwise manner with the Hardy-Littlewood maximal function.

Lemma 2.2.3. Assume that $f \in L^1_{\text{loc}}(\mathbb{R}^n)$. Then

$$\sup_{\epsilon > 0} |(\varphi_\epsilon * f)(x)| \leq Mf(x)$$

for every $x \in \mathbb{R}^n$.

Proof. We begin by considering a function ψ of the form

$$\psi(x) = \sum_{j=1}^N a_j \chi_{B(0, r_j)}(x), \tag{2.6}$$

where $x \in \mathbb{R}^n$, $a_j > 0$ for every $j = 1, \dots, N$. We have that

$$\|\psi\|_{L^1(\mathbb{R}^n)} = \sum_{j=1}^N a_j |B(0, r_j)|.$$

Let $x \in \mathbb{R}^n$ and $\epsilon > 0$. A change of variables $z = \frac{y}{\epsilon}$ gives us

$$\begin{aligned} |(\psi_\epsilon * f)| &= \left| \int_{\mathbb{R}^n} \psi_\epsilon(y) f(x - y) dy \right| = \left| \frac{1}{\epsilon^n} \int_{\mathbb{R}^n} \psi\left(\frac{y}{\epsilon}\right) f(x - y) dy \right| \\ &= \left| \int_{\mathbb{R}^n} \psi(z) f(x - \epsilon z) dz \right| = \left| \sum_{j=1}^N a_j \int_{B(0, r_j)} f(x - \epsilon z) dz \right| \\ &\leq \sum_{j=1}^N a_j \frac{|B(0, r_j)|}{|B(0, r_j)|} \int_{B(0, r_j)} |f(x - \epsilon z)| dz. \end{aligned}$$

We perform another change of variables, $v = x - \epsilon z$, on the last integral in the inequality

above,

$$\begin{aligned}
\frac{1}{|B(0, r_j)|} \int_{B(0, r_j)} |f(x - \epsilon z)| dz &= \frac{1}{\epsilon^n |B(0, r_j)|} \int_{B(x, \epsilon r_j)} |f(v)| dv \\
&= \frac{1}{|B(x, \epsilon r_j)|} \int_{B(x, \epsilon r_j)} |f(v)| dv \\
&\leq Mf(x).
\end{aligned}$$

With that we conclude,

$$|(\psi_\epsilon * f)| \leq \sum a_j |B(0, r_j)| Mf(x) = \|\psi\|_{L^1(\mathbb{R}^n)} Mf(x)$$

for every $x \in \mathbb{R}^n$. We now consider an increasing sequence of functions ψ_k such that $\psi_k \rightarrow \varphi$ for every $x \in \mathbb{R}^n$ as $k \rightarrow \infty$. This sequence consists of sums of characteristic functions, 2.6, that depend on open balls centered at zero. Therefore, we can choose constants a_j to sum towards the value of $\varphi(x)$ as we increase the sequence for every $x \in \mathbb{R}^n$. By the monotone convergence theorem,

$$\begin{aligned}
|(\varphi_\epsilon * f)(x)| &\leq \int_{\mathbb{R}^n} \varphi_\epsilon(y) |f(x - y)| dy = \int_{\mathbb{R}^n} \lim_{k \rightarrow \infty} (\psi_k)_\epsilon(y) |f(x - y)| dy \\
&= \lim_{k \rightarrow \infty} \int_{\mathbb{R}^n} (\psi_k)_\epsilon(y) |f(x - y)| dy = \lim_{k \rightarrow \infty} |((\psi_k)_\epsilon * |f|)(x)| \\
&\leq \lim_{k \rightarrow \infty} \|\psi_k\|_{L^1(\mathbb{R}^n)} Mf(x) = \|\varphi\|_{L^1(\mathbb{R}^n)} Mf(x) = Mf(x)
\end{aligned}$$

for every $\epsilon > 0$ and $x \in \mathbb{R}^n$. □

An application of the Hardy-Littlewood maximal function is illustrated in the proof for the following lemma. The lemma focuses on the density of $C^\infty(\mathbb{R}^n)$ in $W^{1,p}(\mathbb{R}^n)$, however, the proof also shows that $C^\infty(\mathbb{R}^n)$ is dense in $L^p(\mathbb{R}^n)$ for $1 < p < \infty$. More specifically, we have that the convolutions of functions in $L^p(\mathbb{R}^n)$ get arbitrarily close to the functions themselves.

Lemma 2.2.4. Let $1 < p < \infty$ and assume that $u \in W^{1,p}(\mathbb{R}^n)$. Then $\varphi_\epsilon * u \rightarrow u$ in $W^{1,p}(\mathbb{R}^n)$ as $\epsilon \rightarrow 0$. More specifically, the space $C^\infty(\mathbb{R}^n) \cap W^{1,p}(\mathbb{R}^n)$ is dense in $W^{1,p}(\mathbb{R}^n)$.

Proof. Let $\epsilon > 0$ and $u \in L^p(\mathbb{R}^n)$ with

$$u(x) = u^*(x) = \lim_{r \rightarrow 0} \frac{1}{|B(x, r)|} \int_{B(x, r)} u(y) dy$$

and let x be a Lebesgue point of u . We have that,

$$\begin{aligned} |(\varphi_\epsilon * u)(x) - u(x)| &= \left| \int_{\mathbb{R}^n} \varphi_\epsilon(y) (u(x-y) - u(x)) dy \right| \\ &\leq \frac{1}{\epsilon^n} \int_{B(0, \epsilon)} \varphi\left(\frac{y}{\epsilon}\right) |u(x-y) - u(x)| dy \\ &\leq \frac{\|\varphi\|_{L^\infty(\mathbb{R}^n)}}{\epsilon^n} \int_{B(0, \epsilon)} |u(x-y) - u(x)| dy. \end{aligned}$$

Performing a change of variables, with $z = x - y$, we have

$$\frac{\|\varphi\|_{L^\infty(\mathbb{R}^n)}}{\epsilon^n} \int_{B(0, \epsilon)} |u(x-y) - u(x)| dy = \frac{C(n)\|\varphi\|_{L^\infty(\mathbb{R}^n)}}{|B(x, \epsilon)|} \int_{B(x, \epsilon)} |u(z) - u(x)| dz,$$

where $C(n)$ is a constant dependent on n . Since x is a Lebesgue point,

$$|(\varphi_\epsilon * u)(x) - u(x)| \leq \frac{C(n)\|\varphi\|_{L^\infty(\mathbb{R}^n)}}{|B(x, \epsilon)|} \int_{B(x, \epsilon)} |u(z) - u(x)| dz \rightarrow 0$$

as $\epsilon \rightarrow 0$. By lemma 2.2.3,

$$|(\varphi_\epsilon * u)(x) - u(x)| \leq |(\varphi_\epsilon * u)(x)| + |u(x)| \leq Mu(x) + |u(x)|.$$

Theorem 1.2.4 implies that almost every point of $x \in \mathbb{R}^n$ is a Lebesgue point. Therefore, by Theorem 1.1.10(c), we have that $Mu + |u| \in L^p(\mathbb{R}^n)$, and by the dominated convergence theorem,

$$\lim_{\epsilon \rightarrow 0} \int_{\mathbb{R}^n} |(\varphi * u)(x) - u(x)|^p dx = \int_{\mathbb{R}^n} \lim_{\epsilon \rightarrow 0} |(\varphi * u)(x) - u(x)|^p dx = 0.$$

□

Recall that a function is L -Lipschitz continuous if for a set $A \in \mathbb{R}^n$ and $0 \leq L < \infty$ with $u : A \rightarrow \mathbb{R}$, we have that

$$|u(x) - u(y)| \leq L|x - y|$$

for every $x, y \in A$. The space of all Lipschitz functions in A is denoted by $\text{Lip}(A)$. We have the following results for Lipschitz functions.

Lemma 2.2.5. Let $\Omega \subset \mathbb{R}^n$ be an open set and assume that $u \in \text{Lip}(\Omega)$. Then u is differentiable almost everywhere in Ω .

We are often interested in $|\nabla u|$ instead of ∇u , for which the following lemma is useful.

Lemma 2.2.6. Let $\Omega \subset \mathbb{R}^n$ be an open set and assume that $u \in \text{Lip}(\Omega)$. Then

$$|\nabla u(x)| = \lim_{r \rightarrow 0} \sup_{y \in B(x, r)} \frac{|u(y) - u(x)|}{r} \quad (2.7)$$

for almost every $x \in \Omega$. Note that since $u \in \text{Lip}(\Omega)$, we have that $|\nabla u(x)| \leq L$ for almost every $x \in \Omega$.

An interesting application of convolution approximation 2.5, Lemmas 2.2.4, 2.2.5 and 2.2.6, allows us to identify the space of Lipschitz continuous functions with the Sobolev space $W^{1, \infty}(\mathbb{R}^n)$.

Theorem 2.2.7. Assume that $u \in L^\infty(\mathbb{R}^n)$. Then the following hold:

1. $u \in W^{1, \infty}(\mathbb{R}^n)$ if and only if u has a representative in $\text{Lip}(\mathbb{R}^n)$.
2. If $u \in W^{1, \infty}(\mathbb{R}^n)$, then there exists a set $E \subseteq \mathbb{R}^n$ such that $|E| = 0$ and

$$|u(x) - u(y)| \leq \|\nabla u\|_{L^\infty(\mathbb{R}^n)} |x - y|$$

for every $x, y \in \mathbb{R}^n \setminus E$.

3. If u is L -Lipschitz in $\mathbb{R}^n$, then $\|\nabla u\|_{L^\infty(\mathbb{R}^n)} \leq L$.

Proof. Assume that $u \in W^{1, \infty}(\mathbb{R}^n)$ and define the convolution (2.5) $u_\epsilon = \varphi_\epsilon * u$ for every $\epsilon > 0$. Let $\epsilon > 0$. By Young's inequality (2.3),

$$\|\nabla u_\epsilon\|_{L^\infty(\mathbb{R}^n)} = \|\varphi_\epsilon * \nabla u\|_{L^\infty(\mathbb{R}^n)} \leq \|\varphi_\epsilon\|_{L^\infty(\mathbb{R}^n)} \|\nabla u\|_{L^\infty(\mathbb{R}^n)} \leq \|\nabla u\|_{L^\infty(\mathbb{R}^n)},$$

where the first convolution is taken componentwise. Since $u_\epsilon \in C^\infty(\mathbb{R}^n)$, the Fundamental Theorem of Calculus gives

$$\begin{aligned} |u_\epsilon(x) - u_\epsilon(y)| &= \left| \int_0^1 \nabla u_\epsilon(y + t(x - y)) \cdot (x - y) dt \right| \\ &\leq \|\nabla u_\epsilon\|_{L^\infty(\mathbb{R}^n)} |x - y| \leq \|\nabla u\|_{L^\infty(\mathbb{R}^n)} |x - y| \end{aligned}$$

for every $x, y \in \mathbb{R}^n$. The proof of Lemma 2.2.4 shows that the convolutions u_ϵ converge to u pointwise almost everywhere in $\mathbb{R}^n$ as $\epsilon \rightarrow 0$. Hence, there is a set $E \subset \mathbb{R}^n$ such that $|E| = 0$ and

$$|u(x) - u(y)| \leq \|\nabla u\|_{L^\infty(\mathbb{R}^n)} |x - y| \quad (2.8)$$

for every $x, y \in \mathbb{R}^n \setminus E$. This proves (2).

By uniform continuity, u can be redefined on the set E of measure zero in such a way that (2.8) holds for every $x, y \in \mathbb{R}^n$. Thus, u has a representative in $\text{Lip}(\mathbb{R}^n)$. Now, if we assume that $u \in L^\infty(\mathbb{R}^n)$ and is L -Lipschitz in $\mathbb{R}^n$, then integration by parts used in Definition 2.1.1 tells us that the classical gradient of u , which exists almost everywhere, is the weak gradient of u . By Lemmas 2.2.5 and 2.2.6, we have that $|\nabla u(x)| \leq L$ for almost every $x \in \mathbb{R}^n$. Thus, $\|\nabla u\|_{L^\infty(\mathbb{R}^n)} \leq L$ and $u \in W^{1,\infty}(\mathbb{R}^n)$, proving the remaining assertions. $\square$

2.3 Functions of Bounded Mean Oscillations (BMO)

This section introduces the space of functions of Bounded Mean Oscillation (BMO). This space consists of functions whose mean oscillation is uniformly bounded over all cubes in $\mathbb{R}^n$; that is, the maximal average deviation from the mean over cubes is finite. We also introduce the sharp BMO space, which is defined in terms of the restricted sharp maximal function. The results presented in this section are drawn from [5, Chapter 9].

Definition 2.3.1. A function $f \in L^1_{\text{loc}}(\mathbb{R}^n)$ belongs to the space of Bounded Mean Oscillation, denoted $\text{BMO}(\mathbb{R}^n)$, if

$$\|f\|_{\text{BMO}(\mathbb{R}^n)} = \sup_Q \frac{1}{|Q|} \int_Q |f(x) - f_Q| dx < \infty$$

where the supremum is taken over all cubes $Q \subset \mathbb{R}^n$.

We have that $L^\infty(\mathbb{R}^n) \subset \text{BMO}(\mathbb{R}^n)$ and

$$\|f\|_{\text{BMO}(\mathbb{R}^n)} \leq 2 \|f\|_{L^\infty(\mathbb{R}^n)}.$$

This follows since

$$\begin{aligned} \frac{1}{|Q|} \int_Q |f(x) - f_Q| dx &\leq \frac{1}{|Q|} \int_Q (|f(x)| + |f_Q|) dx = \frac{1}{|Q|} \int_Q |f(x)| dx + |f_Q| \\ &\leq \frac{1}{|Q|} \int_Q |f(x)| dx + |f|_Q \leq \frac{2}{|Q|} \int_Q |f(x)| dx \\ &\leq 2 \|f\|_{L^\infty(\mathbb{R}^n)}. \end{aligned}$$

We have that $L^\infty(\mathbb{R}^n)$ is a proper subset of $\text{BMO}(\mathbb{R}^n)$, since the function $\ln|x| \in \text{BMO}(\mathbb{R}^n)$, however, $\ln|x| \notin L^\infty(\mathbb{R}^n)$. See [5], example 9.36.

We now define the space $\text{BMO}(Q_0)$, where Q_0 is a cube in $\mathbb{R}^n$ using the restricted sharp maximal function.

Definition 2.3.2. Let $Q_0 \subset \mathbb{R}^n$ be a cube. A function $f \in L^1(Q_0)$ belongs to the space $\text{BMO}^\#(Q_0)$ if

$$\|f\|_{\text{BMO}^\#(Q_0)} = \sup_{\sigma Q \subset Q_0} \frac{1}{|Q|} \int_Q |f(y) - f_Q| dy < \infty.$$

where the supremum is taken over all cubes such that $\sigma Q \subset Q_0$ and $\sigma \geq 1$.

Note that

$$\sup_{\sigma Q \subset Q_0} \frac{1}{|Q|} \int_Q |f(y) - f_Q| dy = \|M_{\text{res}, Q_0, \sigma}^\# f\|_{L^\infty(Q_0)},$$

therefore, $f \in \text{BMO}^\#(Q_0)$ if and only if $M_{\text{res}, Q_0, \sigma}^\# f \in L^\infty(\mathbb{R}^n)$.

The following lemma is proved in [5, Chapter 9.1]. Recall that $l(Q)$ is the side length of the cube $Q \in \mathbb{R}^n$.

Lemma 2.3.3. Let $Q \subset \mathbb{R}^n$ be a cube. Then there exists a constant $C = C(n)$ such that,

$$\int_Q |u(x) - u_Q| dx \leq C l(Q) \int_Q |\nabla u(x)| dx$$

for every $u \in W^{1,1}(Q)$.

With this lemma, we are able to identify functions in $\text{BMO}^\#(Q_0)$ with functions that are in the first order Sobolev space $W^{1,n}(Q_0)$ where $Q_0 \subset \mathbb{R}^n$.

Corollary 2.3.4. Let $Q_0 \subset \mathbb{R}^n$ be a cube and let $u \in W^{1,n}(Q_0)$. Then $u \in \text{BMO}^\#(Q_0)$.

Proof. By Hölder's inequality, we have that $u \in W^{1,1}$. By Lemma 2.3.3 and Hölder's inequality, we have that

$$\begin{aligned} \frac{1}{|Q|} \int_Q |u(x) - u_Q| dx &\leq C(n) l(Q) \left(\frac{1}{|Q|} \int_Q |\nabla u(x)|^n dx \right)^{\frac{1}{n}} \\ &= C(n) l(Q) |Q|^{-\frac{1}{n}} \left(\int_Q |\nabla u(x)|^n dx \right)^{\frac{1}{n}}. \end{aligned}$$

for every Q such that $\sigma Q \subset Q_0$. Since $l(Q)|Q|^{-\frac{1}{n}} = 1$ and $Q \subset Q_0$, we have

$$\begin{aligned} \|u\|_{\text{BMO}^\#(Q_0)} &= \sup_{\sigma Q \subset Q_0} \frac{1}{|Q|} \int_Q |u(x) - u_Q| dx \\ &\leq C(n) \left(\int_Q |\nabla u(x)|^n dx \right)^{\frac{1}{n}} \\ &\leq C(n) \|u\|_{W^{1,n}(Q_0)}. \end{aligned}$$

□

Chapter 3

The local Fefferman-Stein inequality on a cuspidal domain

This chapter is dedicated to establishing a local Fefferman–Stein inequality on a specific cuspidal (irregular) domain in $\mathbb{R}^2$. The Fefferman–Stein inequality was first introduced by Charles Fefferman and Elias M. Stein in [3]. A local version of the classical Fefferman–Stein inequality was originally introduced by Diening, Růžička, and Schumacher in [2], where it was proven for bounded John domains, which are regular in the geometric sense. However,

In contrast, the domain of interest in this chapter (see (3.9)), is not a John domain, and we demonstrate that the unweighted local Fefferman–Stein inequality fails to hold on such a domain. Our argument relies on the fact that the Fefferman–Stein inequality implies the Poincaré inequality; since the Poincaré inequality is known to fail on non-John domains, this establishes the failure of the Fefferman–Stein inequality in the cuspidal setting.

Despite this, we prove a weaker, weighted version of the Fefferman–Stein inequality that does hold on the irregular domain. Our approach is based on a decomposition technique introduced in [1], which was originally developed to study the existence of solutions to the divergence equation on the same domain.

We also refer to Section 5.4 in [9], where the validity of a weighted version of the local Fefferman–Stein inequality is proved on bounded John domains (see Theorem 5.9). The weights considered in the cited manuscript are powers of the distance to the boundary of the domain and the exponent on the weights depend on the Assouad dimension of $\partial\Omega$.

In Section 3.1, we investigate the constant that appears in the Fefferman–Stein inequality on certain regular cubes and other regular domains of interest, focusing on how this constant relates to domain geometry. Section 3.2 recalls the definition of John domains and presents several examples. Section 3.3 reviews results on a specific decomposition of functions, as developed in [1], and its connection to the discrete Hardy inequality. Finally, in Section 3.4, we develop the weighted version of the Fefferman–Stein inequality for a cuspidal domain introduced in (3.9), and show that the Fefferman–Stein inequality does not hold on this domain.

3.1 The Calderon-Zygmund Lemma and the local Fefferman-Stein Inequality on cubes

In this section, we develop key tools used to estimate the local Hardy–Littlewood maximal operator in terms of the local sharp maximal operator. These results are foundational in the study of local Fefferman–Stein inequalities and are particularly relevant in the context of weighted estimates on irregular domains.

We begin with an interesting result named the Calderón–Zygmund decomposition lemma, which identifies regions where the average of a function exceeds a fixed threshold and provides a collection of disjoint subcubes where the function is locally large. This decomposition allows us to locally control the restricted maximal function and supports the derivation of sharp integral estimates.

Let $Q_0 \subset \mathbb{R}^n$ be a cube. We define by induction the families M_k , $k = 0, 1, 2, \dots$ of open

subcubes of the cube Q_0 where $M_0 = \{Q_0\}$. Suppose that the family M_k is given. Then we divide dyadically every cube of M_k into 2^n equal cubes whose interiors are pairwise disjoint. They form together the family M_{k+1} . In general, every two cubes from $M = \bigcup_{k=0} M_k$ are either disjoint or one includes the other. We refer to M as the *dyadic decomposition* of Q_0 . Moreover, any cube Q initiates uniquely the increasing sequence of cubes from M such that

$$Q = Q_k \subset Q_{k-1} \subset \dots \subset Q_1 \subset Q_0,$$

where $Q_s \in M_s$, with $0 \leq s \leq k$. It is worth noting that for Q_k in that sequence, $|Q_k| = 2^{-n}|Q_{k-1}|$. With these families of cubes, we prove the following Calderon-Zygmund lemma.

Lemma 3.1.1. (Calderon-Zygmund lemma) Let $Q_0 \subset \mathbb{R}^n$, and $f \in L^1(Q_0)$. Then, for any $t \geq |f|_{Q_0}$, there exists a collection of disjoint cubes $Q_j^t \in M$ with $j = 1, 2, \dots$, such that

- (a) $t < \frac{1}{|Q|} \int_Q |f(y)| dy \leq 2^n t$ for every cube $Q \in \{Q_j^t\}$;
- (b) if $f_{Q_0} \leq \beta \leq t$, then each Q is a subcube of one $Q_\beta \in \{Q_j^\beta\}$ where $\{Q_j^\beta\}$ is a collection of disjoint cubes in M ;
- (c) $|f(x)| \leq t$ for almost every $x \in Q_0 \setminus \bigcup_j Q_j^t$;
- (d) $\sum_j |Q_j^t| \leq |\{x \in Q_0 : M_{res, Q_0, 1} f(x) > t\}|$;
- (e) $|\{x \in Q_0 : M_{res, Q_0, 1} f(x) > 5^n t\}| \leq 5^n \sum_j |Q_j^t|$.

Proof. Let Q be an arbitrary cube from $M = \bigcup_{k=0} M_k$. We analyze the sequence

$$Q = Q_k \subset Q_{k-1} \subset \dots \subset Q_1 \subset Q_0$$

where $Q_s \in M_s$, and $s = 0, 1, \dots, k$. We include the cube Q_k in the family $\{Q_j^t : j = 0, 1, 2, \dots\}$ if and only if

$$\frac{1}{|Q_k|} \int_{Q_k} |f(y)| dy > t \quad \text{and} \quad \frac{1}{|Q_s|} \int_{Q_s} |f(y)| dy \leq t,$$

for $s = 0, 1, \dots, k-1$. We have that every two cubes Q' and Q'' in $\{Q_j^t : j = 0, 1, 2, \dots\}$ are disjoint because if $Q' \subset Q''$, then we have

$$\frac{1}{|Q''|} \int_{Q''} |f(y)| dy > t \quad \text{and} \quad \frac{1}{|Q''|} \int_{Q''} |f(y)| dy \leq t,$$

which is a contradiction. Since $|f|_{Q_0} \leq t$, $Q_0 \notin \{Q_j^t : j = 0, 1, 2, \dots\}$, and for a cube $Q = Q_k \in \{Q_j^t\}$, we have that $|Q_k| = \frac{1}{2^n} |Q_{k-1}|$ in that sequence, therefore,

$$t < \frac{1}{|Q_k|} \int_{Q_k} |f(y)| dy \leq \frac{1}{|Q_k|} \int_{Q_{k-1}} |f(y)| dy = \frac{2^n}{|Q_{k-1}|} \int_{Q_k} |f(y)| dy \leq 2^n t.$$

This proves assertion (a).

Let $f_{Q_0} \leq \beta \leq t$ and $Q_k \in \{Q_j^t : j = 0, 1, 2, \dots\}$. Then we consider the smallest index s' in the sequence

$$Q = Q_k \subset Q_{k-1} \subset \dots \subset Q_1 \subset Q_0,$$

where $s = 1, \dots, k-1$, for which $\beta < \frac{1}{|Q_{s'}|} \int_{Q_{s'}} |f(y)| dy$. We have that $Q_k \subset Q_{s'} \in \{Q_j^\beta : j = 0, 1, 2, \dots\}$. This proves assertion (b).

Consider $x \notin \bigcup_j Q_j^t$ and let $Q_l \in M$ be a cube containing x . Analyzing

$$Q_l \subset Q_{l-1} \subset \dots \subset Q_1 \subset Q_0$$

we have that none of the cubes in that sequence are in $\{Q_j^t : j = 0, 1, 2, \dots\}$ which means that $\frac{1}{|Q_s|} \int_{Q_s} |f| \leq t$ for any $s = 0, 1, \dots, l$, and therefore, $\frac{1}{|Q_l|} \int_{Q_l} |f| \leq t$. By the Lebesgue Differentiation Theorem, we have that

$$|f(x)| \leq \sup_{Q \ni x : Q \subset M} \frac{1}{|Q|} \int_Q |f(y)| dy \leq t$$

for almost every $x \in Q_0 \setminus \bigcup_j Q_j^t$. This proves assertion (c).

The preceding assertion shows that

$$\bigcup_j Q_j^t = \{x \in Q_0 : \sup_{Q \ni x : Q \subset M} \frac{1}{|Q|} \int_Q |f(y)| dy > t.\}$$

Since

$$\sup_{Q \ni x: Q \subset M} \frac{1}{|Q|} \int_Q |f(y)| dy \leq M_{res, Q_0, 1} f(x),$$

we have that

$$\sum_j |Q_j^t| \leq |\{x \in Q_0 : M_{res, Q_0, 1} f(x) > t.\}|.$$

This proves assertion (d).

Let $x \notin \bigcup_j 5Q_j^t$ and Q be an arbitrary cube such that $x \in Q \subset Q_0$. From assertion (a) and (c), we have that

$$\begin{aligned} \int_Q |f(y)| dy &= \int_{Q \setminus \bigcup_j Q_j^t} |f(y)| dy + \sum_j \int_{Q \cap Q_j^t} |f(y)| dy \\ &\leq t|Q| + \sum_{Q \cap Q_j^t \neq \emptyset} \int_{Q_j^t} |f(y)| dy \\ &\leq t|Q| + 2^n t \sum_{Q \cap Q_j^t \neq \emptyset} |Q_j^t|. \end{aligned}$$

It can be verified that if $Q \not\subset 5Q_j^t$, and $Q \cap Q_j^t \neq \emptyset$, then $Q_j^t \subset 2Q$. We can write,

$$\int_Q |f(y)| dy \leq t|Q| + 2^n t \sum_{Q_j^t \subset 2Q} |Q_j^t| \leq t|Q| + 2^n t |2Q| \leq 5^n t |Q|.$$

Therefore,

$$\frac{1}{|Q|} \int_Q |f(y)| dy \leq 5^n t$$

whenever $x \in Q \setminus \bigcup_j 5Q_j^t$. Moreover, $M_{res, Q_0, 1} f(x) \leq 5^n t$ whenever $x \notin \bigcup_j 5Q_j^t$. This tells us that

$$\{x \in Q_0 : M_{res, Q_0, 1} f(x) > 5^n t\} \subset \bigcup_j 5Q_j^t.$$

This proves assertion (e). □

We now present a result due to Iwaniec [4], which provides an L^p -bound for the local maximal function in terms of the local sharp maximal function and the average of the original function. This inequality plays a crucial role in transferring oscillation control to maximal control, which is a central theme in real-variable harmonic analysis.

Lemma 3.1.2. Let $Q_0 \subset \mathbb{R}^n$ be a cube, $1 \leq p < \infty$, and $f \in L^1(Q_0)$. Suppose $M_{res,Q_0,1}^\# f \in L^p(Q_0)$, then $M_{res,Q_0,1} f \in L^p(Q_0)$ and

$$\begin{aligned} & \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res,Q_0,1} f)(x)|^p dx \right)^{\frac{1}{p}} \\ & \leq 10^{5np} \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res,Q_0,1}^\# f)(x)|^p dx \right)^{\frac{1}{p}} + 10^{n+1} \frac{1}{|Q_0|} \int_{Q_0} |f(x)| dx. \end{aligned} \quad (3.1)$$

For the proof below, we use the notation $\mathcal{G}(Q_0)$ and $\mathcal{D}(Q_0)$ to denote two different dyadic decompositions of Q_0 .

Proof. Let $t \geq 2^{n+1}|f|_{Q_0}$ and $a > 0$. We also let

$$B = \{x \in Q_0 : M_{res,Q_0,1} f > 5^n t\},$$

$$C = \{x \in Q_0 : M_{res,Q_0,1}^\# f > \frac{t}{a}\},$$

and

$$D = \{x \in Q_0 : M_{res,Q_0,1} f > 2^{-n-1} t\}.$$

Our first step will be to verify the following estimation,

$$|B| \leq 5^n |C| + \frac{2 \cdot 5^n}{a} |D|. \quad (3.2)$$

By Lemma 3.1.1 part (e), there exists a collection $\mathcal{G}_\alpha$ of disjoint cubes $Q^\alpha \in \mathcal{G}(Q_0)$ with $j = 1, 2, \dots$, such that

$$|B| = |\{x \in Q_0 : M_{res,Q_0,1} f(x) > 5^n t\}| \leq 5^n \sum_{Q^\alpha \in \mathcal{G}_\alpha} |Q^\alpha|.$$

We have that $2^{-n-1} t \geq 2^{-n-1} (2^{n+1} |f|_{Q_0}) = |f|_{Q_0}$. Thus, by part (d) of Lemma 3.1.1, there exists a collection $\mathcal{F}_\omega$ of disjoint cubes $Q^\omega \in \mathcal{D}(Q_0)$ such that

$$\sum_{Q^\omega \in \mathcal{F}_\omega} |Q^\omega| \leq |\{x \in Q_0 : M_{res,Q_0,1} f(x) > 2^{-n-1} t\}| = |D|.$$

Therefore, to verify (3.2), it suffices to show that

$$\sum_{Q^\alpha \in \mathcal{G}_\alpha} |Q^\alpha| \leq |C| + \frac{2}{a} \sum_{Q^\omega \in \mathcal{F}_\omega} |Q^\omega|. \quad (3.3)$$

We fix a cube $Q^\omega \in \mathcal{F}_\omega$. By part (a) of Lemma 3.1.1,

$$|f|_{Q^\omega} \leq 2^n(2^{-n-1}t) = \frac{t}{2},$$

and for all $Q^\alpha \in \mathcal{G}_\alpha$ we have

$$\int_{Q^\alpha} |f| \geq t|Q^\alpha|.$$

We obtain,

$$\int_{Q^\alpha} |f - f_{Q^\omega}| \geq \left(\int_{Q^\alpha} |f| \right) - |f_{Q^\omega}| |Q^\alpha| \geq t|Q^\alpha| - \frac{t}{2}|Q^\alpha| = \frac{t}{2}|Q^\alpha|.$$

Therefore, the sum over disjoint cubes $Q^\alpha \subseteq Q^\omega$,

$$\sum_{Q^\alpha \subseteq Q^\omega} |Q^\alpha| \leq \frac{2}{t} \sum_{Q^\alpha \subseteq Q^\omega} \int_{Q^\alpha} |f - f_{Q^\omega}| \leq \frac{2}{t} \int_{Q^\omega} |f - f_{Q^\omega}|. \quad (3.4)$$

We now consider two cases:

Case 1: If

$$\frac{1}{|Q^\omega|} \int_{Q^\omega} |f - f_{Q^\omega}| \leq \frac{t}{a},$$

then (3.4) gives

$$\sum_{Q^\alpha \subseteq Q^\omega} |Q^\alpha| \leq \frac{2}{a} |Q^\omega|.$$

Case 2: If

$$\frac{1}{|Q^\omega|} \int_{Q^\omega} |f - f_{Q^\omega}| > \frac{t}{a},$$

then

$$M_{res, Q_0, 1}^\# f(x) > \frac{t}{a}$$

for every $x \in Q^\omega$. Thus, $Q^\omega \subseteq \{x \in Q_0 : M_{res, Q_0, 1}^\# f(x) > \frac{t}{a}\}$ and

$$\sum_{Q^\alpha \subseteq Q^\omega} |Q^\alpha| \leq |Q^\omega \cap \{x \in Q_0 : M_{res, Q_0, 1}^\# f(x) > \frac{t}{a}\}|.$$

Hence, in either cases 1 or 2, we obtain

$$\sum_{Q^\alpha \subseteq Q^\omega} |Q^\alpha| \leq |Q^\omega \cap \{x \in Q_0 : M_{res, Q_0, 1}^\# f(x) > \frac{t}{a}\}| + \frac{2}{a} |Q^\omega|$$

for each cube $Q^\omega \in \mathcal{F}_\omega$. Note that $|f|_{Q_0} \leq 2^{-n-1}t < t$, so by part (b) of Lemma 3.1.1, we have that each disjoint $Q^\alpha \in \mathcal{G}_\alpha$ belongs to one disjoint $Q^\omega \in \mathcal{F}_\omega$. Therefore,

$$\sum_{Q^\alpha \in \mathcal{G}_\alpha} |Q^\alpha| = \sum_{Q^\omega \in \mathcal{F}_\omega} \left(\sum_{Q^\alpha \subseteq Q^\omega} |Q^\alpha| \right) \leq \sum_{Q^\omega \in \mathcal{F}_\omega} \left(|Q^\omega \cap C| + \frac{2}{a} |Q^\omega| \right) \leq |C| + \frac{2}{a} \sum_{Q^\omega \in \mathcal{F}_\omega} |Q^\omega|,$$

which verifies (3.3) and shows that

$$|B| \leq 5^n |C| + \frac{2 \cdot 5^n}{a} |D|.$$

Now, our second step will involve the formula,

$$\int_{Q_0} |F(x)|^p dx = p \int_0^\infty t^{p-1} |\{x \in Q_0 : |F(x)| > t\}| dt \quad (3.5)$$

where $F \in L^p(Q_0)$.

We consider an arbitrary number $N > 2^{n+1}a$ and integrate both sides of (3.2) as

$$p \int_{2^{n+1}|f|_{Q_0}}^N t^{p-1} |B| dt \leq 5^n p \int_{2^{n+1}|f|_{Q_0}}^N t^{p-1} |C| dt + \frac{2 \cdot 5^n \cdot p}{a} \int_{2^{n+1}|f|_{Q_0}}^N t^{p-1} |D| dt.$$

Hence,

$$p \int_{2^{n+1}|f|_{Q_0}}^N t^{p-1} |B| dt \leq 5^n p \int_0^\infty t^{p-1} |C| dt + \frac{2 \cdot 5^n \cdot p}{a} \int_0^\infty t^{p-1} |D| dt.$$

We now perform an appropriate change of variables on each integral to obtain

$$p \int_{2 \cdot 10^n |f|_{Q_0}}^{5^n N} u^{p-1} |E| du \leq 5^{np+p} a^p p \int_0^\infty u^{p-1} |F| du + \frac{2 \cdot 10^{pn+p} \cdot p}{a} \int_0^{5^n N} u^{p-1} |E| du. \quad (3.6)$$

where

$$E = \{x \in Q_0 : M_{res, Q_0, 1} f > u\},$$

and

$$F = \{x \in Q_0 : M_{res, Q_0, 1}^\# f > u\}.$$

Note that the inequality

$$p \int_0^{2 \cdot 10^n |f|_{Q_0}} u^{p-1} |E| du \leq |Q_0| p \int_0^{2 \cdot 10^n |f|_{Q_0}} u^{p-1} du = |Q_0| (2 \cdot 10^n \cdot |f|_{Q_0})^p,$$

together with (3.5) and (3.6), gives

$$p \int_0^{5^n N} u^{p-1} |E| du \leq 5^{np+p} a^p \int_{Q_0} |M_{res, Q_0, 1}^\#|^p + \frac{2 \cdot 10^{pn+p} \cdot p}{a} \int_0^{5^n N} u^{p-1} |E| du + (2 \cdot 10^n \cdot |f|_{Q_0})^p \cdot |Q_0|.$$

We set $a = 4 \cdot 10^{np+p}$ and obtain

$$p \int_0^{5^n N} u^{p-1} |E| du < 5^{np+p} \cdot 4^p \cdot 10^{np^2+p^2} \int_{Q_0} |M_{res, Q_0, 1}^\#|^p + \frac{p}{2} \int_0^{5^n N} u^{p-1} |E| du + 2^p \cdot 10^{np} \cdot |f|_{Q_0}^p \cdot |Q_0|.$$

We subtract the middle term of the right side of that inequality and multiply by 2,

$$p \int_0^{5^n N} u^{p-1} |E| du < 2 \cdot 5^{np+p} \cdot 4^p \cdot 10^{np^2+p^2} \int_{Q_0} |M_{res, Q_0, 1}^\#|^p + 2^{p+1} \cdot 10^{np} \cdot |f|_{Q_0}^p \cdot |Q_0|.$$

Since $M_{res, Q_0, 1}^\# f \in L^p(Q_0)$ and N is arbitrary, then we have that

$$\int_0^\infty u^{p-1} |E| du < \infty.$$

We use (3.5) and obtain

$$\int_{Q_0} |M_{res, Q_0, 1}|^p \leq 2 \cdot 4^p \cdot 10^{np^2+p^2} \int_{Q_0} |M_{res, Q_0, 1}^\#|^p + 2^{p+1} \cdot 10^{np} \cdot |Q_0| \left(\frac{1}{|Q_0|} \int_{Q_0} |f| \right)^p$$

We divide by $|Q_0|$ and obtain

$$\begin{aligned} & \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res, Q_0, 1} f)(x)|^p dx \right)^{\frac{1}{p}} \\ & \leq 10^{5np} \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res, Q_0, 1}^\# f)(x)|^p dx \right)^{\frac{1}{p}} + 10^{n+1} \frac{1}{|Q_0|} \int_{Q_0} |f(x)| dx \end{aligned}$$

which is our desired result. $\square$

As an application of the previous lemma, we now derive a useful consequence for the space $L_0^p(Q_0)$, consisting of integrable functions with vanishing mean value.

Corollary 3.1.3. Let $Q_0 \subset \mathbb{R}^n$ be a cube, $1 \leq p < \infty$, and let $f \in L_0^1(Q_0)$. Suppose $M_{res,Q_0,1}^\# f \in L^p(Q_0)$, then $f \in L_0^p(Q_0)$ and

$$\|f\|_{L_0^p(Q_0)} \leq 10^{5np+n+1} \|M_{res,Q_0,1}^\# f\|_{L^p(Q_0)}.$$

Proof. Since $f_{Q_0} = 0$, we have

$$\frac{1}{|Q_0|} \int_{Q_0} |f(y)| dy \leq M_{res,Q_0,1}^\# f(x)$$

for all $x \in Q_0$. We take the mean value and obtain,

$$\frac{1}{|Q_0|} \int_{Q_0} |f(y)| dy \leq \frac{1}{|Q_0|} \int_{Q_0} |M_{res,Q_0,1}^\# f(x)| dx.$$

From Hölder's Inequality, we have

$$\frac{1}{|Q_0|} \int_{Q_0} |M_{res,Q_0,1}^\# f(x)| dx \leq \left(\frac{1}{|Q_0|} \int_{Q_0} |M_{res,Q_0,1}^\# f(x)|^p dx \right)^{\frac{1}{p}}.$$

Therefore,

$$\frac{1}{|Q_0|} \int_{Q_0} |f(y)| dy \leq \left(\frac{1}{|Q_0|} \int_{Q_0} |M_{res,Q_0,1}^\# f(x)|^p dx \right)^{\frac{1}{p}}$$

and by Lemma 3.1.2,

$$\begin{aligned} & \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res,Q_0,1} f)(x)|^p dx \right)^{\frac{1}{p}} \\ & \leq 10^{5np} \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res,Q_0,1}^\# f)(x)|^p dx \right)^{\frac{1}{p}} + 10^{n+1} \frac{1}{|Q_0|} \int_{Q_0} |f(x)| dx \\ & \leq 10^{5np+n+1} \left(\frac{1}{|Q_0|} \int_{Q_0} |(M_{res,Q_0,1}^\# f)(x)|^p dx \right)^{\frac{1}{p}}. \end{aligned}$$

We then multiply both sides by $|Q_0|^{\frac{1}{p}}$ and use Corollary 1.3.2 to obtain

$$|f(x)| \leq 10^{5np+n+1} \left(\int_{Q_0} |(M_{res,Q_0,1}^\# f)(x)|^p dx \right)^{\frac{1}{p}}.$$

for almost every $x \in Q_0$. This proves our desired result. □

Remark 3.1.4. Corollary 3.1.3 can be generalized to a domain $\Omega \subset \mathbb{R}^n$, and $\sigma \geq 1$, instead of $\sigma = 1$. Indeed, there exists a constant $C > 0$ such that

$$\|f\|_{L_0^p(\Omega)} \leq C \|M_{res, \Omega, \sigma}^\# f\|_{L^p(\Omega)}, \quad (3.7)$$

for any $f \in L_0^1(\Omega)$, where the constant C depends only on n, p , and σ . This result is stated in [2, Remark 5.28].

3.2 John Domains

The largest class of domains where the validity of the local Fefferman-Stein inequality has been proved (we refer to [2]) is the class of John domains. Later in this chapter, we study the validity of this inequality on a cuspidal domain (3.9) in $\mathbb{R}^2$. This domain is not John, thus the results proved by Diening et. al. do not hold in this case. In this direction, we prove a weaker weighted version of the local Fefferman-Stein inequality on this irregular domain introduced below.

Let us define the class of John domains and exhibit a few examples.

Definition 3.2.1. A bounded domain $\Omega \subset \mathbb{R}^n$ is a John domain if there exists a point $x_0 \in \Omega$ (referred to as the John center) and a constant $C > 0$, such that for every point $x \in \Omega$ there exists a rectifiable path parametrized by arc length $\varphi : [0, l(\varphi)] \rightarrow \mathbb{R}^n$, that connects x to x_0 , where $\varphi(0) = x$, $\varphi(l(\varphi)) = x_0$ and

$$d(\varphi(t), \partial\Omega) \geq \frac{1}{C} l(\varphi(t), x_0) \quad (3.8)$$

where $l(\varphi(t), x_0)$ is the arc length from $\varphi(t)$ to the point x_0 .

Some examples of John domains include any convex domain in $\mathbb{R}^n$ and Koch's snowflake (see Figure 3.1).

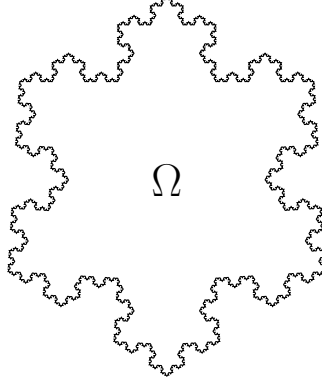


Figure 3.1: Koch's Snowflake is a John domain for any iteration.

The set

$$\Omega := \{(x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1 \text{ and } 0 < x_2 < x_1^\gamma\}$$

is a John domain if $\gamma = 1$. For $\gamma > 1$, we have a cusp near the origin (See Figure 3.2).

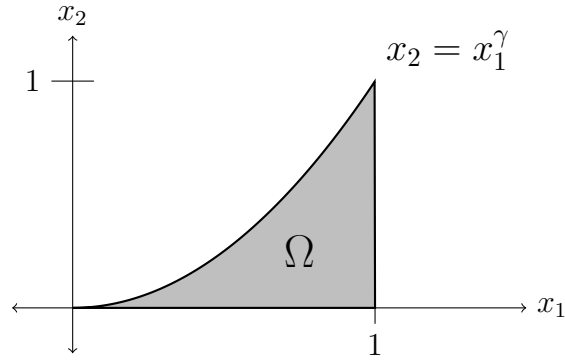


Figure 3.2: A domain that is not John if $\gamma > 1$

Points around this cusp will invalidate inequality (3.8), because points close to the origin will have paths connecting to the John center with greater length relative to the distance from the boundary.

3.3 $\mathcal{C}$ -orthogonal decomposition

In this section, we present a decomposition technique adapted to the singularity of the cuspidal domain introduced in (3.9). The results discussed here are developed in [1] and provide a foundational tool for our analysis in Section 3.4, where we establish a weighted version of the local Fefferman–Stein inequality.

To address the singularity near the cusp, we decompose the domain Ω into dyadic subdomains $\{\Omega_i\}_{i \geq 0}$, as defined in (3.10). This decomposition allows us to work locally while still controlling global behavior through appropriately chosen weights. In particular, we introduce a class of weights, called *locally quasi-constant*, that are compatible with this partition and capture the geometric degeneracy of the domain.

The main result of this section is a $\mathcal{C}$ -orthogonal decomposition theorem (Theorem 3.3.5), which allows any integrable function with vanishing mean value to be decomposed into pieces supported on the subdomains Ω_i , each with vanishing local mean. The decomposition is stable in weighted L^q norms and plays a key role in proving weighted inequalities on non-smooth domains.

Let us introduce a decomposition of the domain

$$\Omega := \{(x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1 \text{ and } 0 < x_2 < x_1^\gamma\} \quad (3.9)$$

which is

$$\Omega_i := \{(x_1, x_2) \in \Omega : 2^{-(i+2)} < x_1 < 2^{-i}\}. \quad (3.10)$$

This decomposition will be of use to us in Section 3.4, where we will apply weights of the type defined below to establish a weighted Fefferman–Stein Inequality on Ω .

Definition 3.3.1. A weight $\omega : \Omega \rightarrow \mathbb{R}$ is called *locally quasi-constant* if $\omega^p(x) \in L^1(\Omega)$ and there exists a uniform constant C_ω such that

$$\operatorname{ess\,sup}_{x \in \Omega_i} \omega(x) \leq C_\omega \operatorname{ess\,inf}_{x \in \Omega_i} \omega(x), \quad (3.11)$$

for all $i \geq 0$. Notice that locally quasi-constant weights are subordinate to a partition $\{\Omega_i\}_{i \geq 0}$ of Ω introduced in (3.10), and $1 < p < \infty$.

Example 3.3.2. The function $\omega(x) := x_1^\beta$, where $\beta > \frac{-\gamma-1}{p}$, is a locally quasi-constant weight with $C_\omega = 2^{2|\beta_0|}$, where $\beta_0 := \frac{-\gamma-1}{p}$.

We recall the following Hardy-type inequalities. Given $p > 1$, *the discrete Hardy inequality* states

$$\sum_{n=1}^{\infty} \left(\frac{1}{n} \sum_{k=1}^n a_k \right)^p \leq \left(\frac{p}{p-1} \right)^p \sum_{n=1}^{\infty} a_n^p,$$

for any non-negative sequence $\{a_n\}_{n \geq 1}$, where the constant in the inequality $(\frac{p}{p-1})^p$ is optimal. This inequality has been widely studied and many generalizations have been shown. For details concerning the prehistory and history of the Hardy inequality we refer the reader to [6] and [7] respectively. A weighted version of the classical discrete Hardy inequality is:

$$\left(\sum_{n=1}^{\infty} u_n \left(\sum_{k=1}^n a_k \right)^p \right)^{1/p} \leq C \left(\sum_{n=1}^{\infty} v_n a_n^p \right)^{1/p}.$$

We also introduce the following weighted discrete Hardy-type inequalities:

$$\sum_{j=1}^{\infty} |\Omega_j| \omega_j^p \left(\sum_{i=1}^j d_i \right)^p \leq C_H^p \sum_{j=1}^{\infty} |\Omega_j| \omega_j^p d_j^p, \quad (3.12)$$

and

$$\sum_{i=1}^{\infty} |\Omega_i|^{1-q} \omega_i^{-q} \left(\sum_{j=i}^{\infty} b_j \right)^q \leq C_H^q \sum_{i=1}^{\infty} |\Omega_i|^{1-q} \omega_i^{-q} b_i^q. \quad (3.13)$$

where inequality (3.12) is to the power p , the weight of the sequence is $u_n = v_n = |\Omega_n| \omega_n^p$, and the inequality (3.13) is its dual version.

Definition 3.3.3. Given $g : \Omega \rightarrow \mathbb{R}$, an integrable function with vanishing mean value, i.e. $\int_{\Omega} g = 0$, we refer by a *$\mathcal{C}$ -orthogonal decomposition of g* subordinate to $\{\Omega_i\}_{i \geq 0}$, to a collection of integrable functions $\{g_i\}_{i \geq 0}$ with the following properties:

1. $g(x) = \sum_{i \geq 0} g_i(x)$.
2. $\text{supp}(g_i) \subset \Omega_i$, for all $i \geq 0$.
3. $\int_{\Omega_i} g_i = 0$, for all $i \geq 0$.

The letter $\mathcal{C}$ in the previous definition refers to the space of constant functions. Notice that having a vanishing mean value could also be understood as orthogonal to the functions in $\mathcal{C}$.

Remark 3.3.4. Notice that decomposition $\{\Omega_i\}_{i \geq 0}$ of Ω defined in (3.10), and the condition (3.11) for locally quasi-constant weights are natural assumptions to address the singularity of the domain introduced in (3.9) and to consider the weights of our interest, which involve the distance to the cusp. For other domains or weights, the partition of the domain and the conditions on the weights might be different.

The next theorem is proved in [1], and will be useful in Section 3.4.

Theorem 3.3.5. Let $\omega : \Omega \rightarrow \mathbb{R}$ be an locally quasi-constant weight that satisfies (3.12) for the sequence weight $\omega_i := \omega(2^{-i})$. Then, given $g \in L^1(\Omega)$, with $\int_{\Omega} g = 0$, there exists $\{g_t\}_{t \in \Gamma}$, a $\mathcal{C}$ -decomposition of g subordinate to $\{\Omega_i\}_{i \geq 0}$ (see Definition 3.3.3), such that

$$\sum_{i=0}^{\infty} \int_{\Omega_i} |g_i(x)|^q \omega^{-q}(x) dx \leq C_d^q \int_{\Omega} |g(x)|^q \omega^{-q}(x) dx. \quad (3.14)$$

Moreover, we have the following estimate for the optimal constant C_d :

$$C_d \leq 2^{2+1/q} C_{\omega}^2 C_H. \quad (3.15)$$

3.4 The Local Fefferman-Stein Inequality on the cuspidal domain

In this section, we examine the validity the local Fefferman–Stein inequality on an irregular domain (i.e., a domain that is not a John domain). More specifically, we study the inequality

$$\|f(x)\|_{L_0^p(\Omega)} \leq C \|\omega(x) M_{res, \Omega, \sigma}^\# f(x)\|_{L^p(\Omega)},$$

where the domain Ω is

$$\Omega := \{(x_1, x_2) \in \mathbb{R}^2 : 0 < x_1 < 1 \text{ and } 0 < x_2 < x_1^\gamma\}$$

with $\gamma > 1$, and a decomposition of Ω is

$$\Omega_i := \{(x_1, x_2) \in \Omega : 2^{-(i+2)} < x_1 < 2^{-i}\}.$$

Here, $\omega(x)$ is a weight related to the singularity that appears in the domain. Indeed, Theorem 3.4.3 presents a weaker version of the local Fefferman–Stein inequality on a cuspidal domain, since the weighted norm on the right-hand side of the inequality (3.21) is larger than the standard (unweighted) norm.

To begin our analysis, we show that the local Fefferman–Stein inequality remains unchanged under rigid motions. The proof relies on a change of variables argument and the fact that averages are preserved under bijective affine transformations.

Proposition 3.4.1. Let $U \subset \mathbb{R}^2$ be a bounded domain where the Fefferman–Stein inequality holds for $1 \leq p < \infty$ and $\sigma \geq 1$. Namely, there is a constant C_U such that

$$\|f\|_{L_0^p(U)} \leq C_U \|M_{res, U, \sigma}^\# f\|_{L^p(U)} \quad (3.16)$$

for any $f \in L^p(U)$, with mean value zero. Then, given a rigid motion $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, inequality (3.16) holds on $T(U)$ with the same constant. Namely,

$$\|g\|_{L_0^p(T(U))} \leq C_U \|M_{res, T(U), \sigma}^\# g\|_{L^p(T(U))}$$

where $g(T(x)) = f(x)$ for all $x \in U$.

Proof. Since T is a rigid motion we have that any cube in $T(U)$ can be written as $T(Q)$, where Q is a cube in U . Also, T is a bijective affine transformation with constant Jacobian

J_T . Now, let $g \in L^p(T(U))$, with mean value zero. Then, we define $f : U \rightarrow \mathbb{R}$ by $f(x) = g(T(x))$. Notice that f belongs to $L^p(U)$ and also has vanishing mean value. Indeed, the average of g on $T(U)$ equals the average of f on U . We develop this fact on cubes in the following lines:

$$\begin{aligned} g_{T(Q)} &= \frac{1}{|T(Q)|} \int_{T(Q)} g(y) \, dy = \frac{1}{\int_{T(Q)} 1 \, dy} \int_{T(Q)} g(y) \, dy \\ &= \frac{J_T}{J_T |Q|} \int_Q f(x) \, dx \\ &= \frac{1}{|Q|} \int_Q f(x) \, dx = f_Q. \end{aligned}$$

Then, by changing variables we have that

$$\begin{aligned} \frac{1}{|T(Q)|} \int_{T(Q)} |g(y) - g_{T(Q)}| \, dy &= \frac{1}{J_T |Q|} \int_{T(Q)} |f(T^{-1}(y)) - f_Q| \, dy \\ &= \frac{1}{|Q|} \int_{T(Q)} |f(T^{-1}(y)) - f_Q| \frac{1}{J_T} \, dy \\ &= \frac{1}{|Q|} \int_Q |f(x) - f_Q| \, dx. \end{aligned}$$

Therefore, given $y = T(x) \in T(U)$, it follows that

$$\begin{aligned} M_{res, T(U), \sigma}^\# g(y) &= \sup_{T(Q) \ni y: \sigma T(Q) \subset T(U)} \frac{1}{|T(Q)|} \int_{T(Q)} |g(u) - g_{T(Q)}| \, du \\ &= \sup_{Q \ni x: \sigma Q \subset U} \frac{1}{|Q|} \int_Q |f(v) - f_Q| \, dv \\ &= M_{res, U, \sigma}^\# f(x). \end{aligned} \tag{3.17}$$

Now, by our assumptions,

$$\begin{aligned} \int_{T(U)} |g(y)|^p \, dy &= J_T \int_U |f(x)|^p \, dx \leq J_T C_U^p \int_U |M_{res, U, \sigma}^\# f(x)|^p \, dx \\ &= C_U^p \int_U |M_{res, T(U), \sigma}^\# g(T(x))|^p J_T \, dx \\ &= C_U^p \int_{T(U)} |M_{res, T(U), \sigma}^\# g(y)|^p \, dy, \end{aligned}$$

which completes the proof. $\square$

Now, we show an argument that proves that the classical unweighted version of the local Fefferman–Stein inequality fails on cuspidal domains. To this end, we recall the improved Poincaré inequality for functions in the first-order Sobolev space $W^{1,p}(\Omega)$. Indeed, this inequality states that under certain conditions on the domain $\Omega \subset \mathbb{R}^n$ and $1 \leq p < \infty$, there exists a constant $C > 0$ such that

$$\|f - f_\Omega\|_{L^p(\Omega)} \leq C \left(\int_\Omega |\nabla f(x) d_\Omega(x)|^p dx \right)^{\frac{1}{p}} \quad (3.18)$$

for any $f \in W^{1,p}(\Omega)$, where $d_\Omega(x)$ is the distance from x to the $\partial\Omega$. It is known that this inequality holds on John domains and fails on irregular domains such as (3.9). In particular, we can observe that it holds on cubes. Moreover, using the same argument discussed in Lemma 3.4.1 with rigid motions, we can conclude that the constant C for cubes remains the same. Indeed, there exists a constant C such that

$$\int_Q |f(x) - f_Q| dx \leq C \int_Q |\nabla f(x)| d_Q(x) dx \quad (3.19)$$

for any $f \in W^{1,1}(\Omega)$ and any cube $Q \subset \Omega$, where $d_Q(x)$ is the distance from x to ∂Q . This property on cubes, in addition to the continuity of the Hardy-Littlewood maximal operator on non-centered cubes, is used in the following theorem to prove that the Fefferman–Stein inequality implies the improved Poincaré inequality on general bounded domains. In particular, using that (3.18) fails on the irregular domain (3.9), we may conclude that the Fefferman–Stein inequality also fails in (3.9).

Theorem 3.4.2. Let $\Omega \subset \mathbb{R}^n$ be a bounded domain where the Fefferman–Stein inequality holds for $1 < p < \infty$ and $\sigma \geq 1$. Namely, there is a constant C such that

$$\|f - f_\Omega\|_{L^p(\Omega)} \leq C \|M_{res,\Omega,\sigma}^\# f\|_{L^p(\Omega)}$$

for any $f \in W^{1,p}(\Omega)$. Then, the improved Poincaré inequality (3.18) holds on Ω for any $f \in W^{1,p}(\Omega)$.

Proof. Note that $d_Q(x) < d_\Omega(x)$ for any cube $Q \subset \Omega$. By (3.19), we have that

$$\begin{aligned}
M_{res,\Omega,\sigma}^\# f(x) &= \sup_{Q \ni x: \sigma Q \subset \Omega} \frac{1}{|Q|} \int_Q |f(y) - f_Q| \, dy \\
&\leq \sup_{Q \ni x: \sigma Q \subset \Omega} \frac{C}{|Q|} \int_Q |\nabla f(y)| d_Q(y) \, dy \\
&\leq \sup_{Q \ni x: \sigma Q \subset \Omega} \frac{C}{|Q|} \int_Q |\nabla f(y)| d_\Omega(y) \, dy \\
&\leq \sup_{Q \ni x} \frac{C}{|Q|} \int_Q |\nabla f(x)| d_\Omega(y) \, dy = CM^*(|\nabla f|d_\Omega)(x)
\end{aligned}$$

where M^* is the non-centered Hardy-Littlewood maximal operator on cubes. Now, using the pointwise estimation between the centered and non-centered maximal operators $M^*(|\nabla f|d_\Omega) \leq C_n M(|\nabla f|d_\Omega)$, for some constant $C > 0$, that depends only on n , we have

$$\begin{aligned}
\|f - f_\Omega\|_{L^p(\Omega)} &\leq C \|M_{res,\Omega,\sigma}^\# f\|_{L^p(\Omega)} \\
&\leq C \|M^*(|\nabla f|d_\Omega)\|_{L^p(\Omega)} \\
&\leq C \|\nabla f\|_{L^p(\Omega)}
\end{aligned}$$

where the last inequality is due to the continuity of the centered Hardy-Littlewood maximal operator. $\square$

Since the improved Poincaré inequality does not hold on a bounded domain $\Omega \subset \mathbb{R}^n$ that is not a John domain for all $f \in W^{1,p}(\Omega)$, we conclude that the unweighted local Fefferman–Stein inequality is also not valid on such a domain.

We now begin to develop a local-to-global argument on the set Ω introduced in (3.9) by establishing a weighted local Fefferman–Stein inequality on each piece of a decomposition of Ω introduced in (3.10).

Lemma 3.4.3. Let $\Omega_i \subset \Omega$ be the set introduced in (3.10), where $i \geq 0$, and $1 \leq p < \infty$. Given $\sigma \geq 1$ there exists a constant $C > 0$, independent of i , such that

$$\|f\|_{L^p(\Omega_i)} \leq C 2^{i(\gamma-1)} \|M_{res,\Omega_i,\sigma}^\# f\|_{L^p(\Omega_i)}, \quad (3.20)$$

for any $f \in L_0^p(\Omega_i)$.

Proof. Our first attempt of Lemma 3.4.3's proof was based on the argument used in Lemma 3.4.1, where the rigid motion T was replaced by another bijective transformation. Indeed, we considered transformations on the style of $T : Q_i \rightarrow \Omega_i$ defined by its inverse

$$T^{-1}(x_1, x_2) = \left(x_1, \frac{1}{3} \cdot 2^{(i+2)} x_2 x_1^\gamma \right),$$

where Q_i is the cube $(2^{-(i+2)}, 2^{-i}) \times (0, 3 \cdot 2^{-(i+2)})$. The biggest challenge we faced in the estimation appeared in the identity (3.17) where it is required to use two different parameters σ 's. After several weeks of working in this direction without obtaining the results we were looking for, we decided to use another approach, more technical, based on the argument implemented in [9, Section 5.4] with the decomposition described in [8, Section 4]. The sketch of the proof follows the notation and results on decomposition of functions described in the papers mentioned above, where, naturally, we consider a different Whitney-type decomposition of the domain in dyadic cubes. Therefore, we sketch the procedure to decompose the subdomains Ω_i in the following steps:

Step 1: We define a dyadic cube as shown in Figure 3.3, with side length similar to $(2^{-i})^{\gamma-1}$.

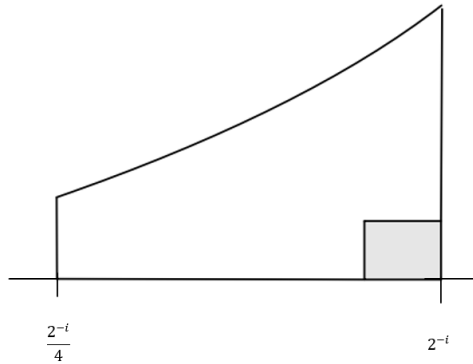


Figure 3.3: A dyadic cube in Ω_i

Step 2: We define a collection of dyadic cubes in the style of Whitney where the side length is comparable to the distance to the graph of the function $x_2 = x_1^\gamma$. It can be seen that in this step we have approximately $(2^{-i})^{\gamma-1}$ cubes with side length similar to $(2^{-i})^\gamma$. See Figure 3.4.

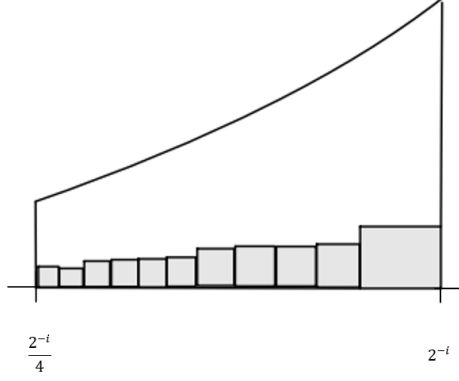


Figure 3.4: A collection of dyadic cubes in Ω_i

Step 3: In this step, we consider a collection of dyadic cubes, also in the style of Whitney, that pile up over the cubes considered in the previous step.

Finally, we define the tree order that considers the cube in Step 1 as the root, the cubes in Step 2 in a total order from right to left, and the parent of each cube in Step 3 as the dyadic cube where it is piling up. See Figure 3.5.

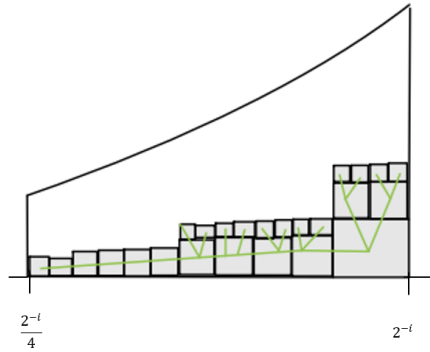


Figure 3.5: A collection of dyadic cubes in Ω_i

Notice that the derivative of the function $x_2 = x_1^\gamma$ is uniformly bounded on $(0, 1)$. Thus, the sum of the measure of all the cubes that pile up on a cube Q in Step 2 is comparable to the measure of Q . This collection of cubes with the order described above has the property that the ratio of the measure of the shadow of a cube over the measure of each cube is uniformly bounded by $C2^{-i(1-\gamma)}$, where C is independent of i .

□

The following two theorems are some of the main results of this thesis, in which we establish a weighted local Fefferman-Stein inequality on the cuspidal domain.

Theorem 3.4.4. Let $\Omega \subset \mathbb{R}^2$ be the set introduced in 3.9 and $1 \leq p < \infty$. Given $\sigma \geq 1$ there exists a constant $C > 0$ such that

$$\|f\|_{L_0^p(\Omega)} \leq C \|d_0^{1-\gamma} M_{res, \Omega, \sigma}^\# f\|_{L^p(\Omega)}, \quad (3.21)$$

for any $f \in L_0^p(\Omega)$, where $d_0(x) = \sqrt{x_1^2 + x_2^2}$.

Proof. Using a dual argument, we consider $f \in L_0^p(\Omega)$ and $g \in L^q(\Omega)$, with $\frac{1}{p} + \frac{1}{q} = 1$ and $\|g\|_{L^q(\Omega)} = 1$. Thus, it is sufficient to show that there exists a constant C , independent of f and g , such that

$$\int_{\Omega} fg \leq C \|d_0^{1-\gamma} M_{res, \Omega, \sigma}^\# f\|_{L^p(\Omega)}.$$

Let us also assume that g integrates zero. Then, using Theorem 3.3.3, we can write g as $g(x) = \sum_{i \geq 0} g_i(x)$. Then, using the bounded overlapping of $\{\Omega_i\}$, dominated convergence, and that each function g_i integrates zero we have

$$\begin{aligned} \int_{\Omega} f(x)g(x) \, dx &= \int_{\Omega} f(x) \left(\sum_{i \geq 0} g_i(x) \right) \, dx = \int_{\Omega} \left(\sum_{i \geq 0} f(x)g_i(x) \right) \, dx \\ &= \sum_{i \geq 0} \int_{\Omega} f(x)g_i(x) \, dx = \sum_{i \geq 0} \int_{\Omega_i} f(x)g_i(x) \, dx \\ &= \sum_{i \geq 0} \int_{\Omega_i} (f(x) - f_{\Omega_i}) g_i(x) \, dx. \end{aligned}$$

We continue by using the Hölder inequality in the integral, the local Fefferman-Stein inequality stated in Lemma 3.4.3, and the monotonicity of the sharp maximal operator it follows that

$$\begin{aligned}
\int_{\Omega} f(x)g(x) \, dx &\leq \sum_{i \geq 0} \|f - f_{\Omega_i}\|_{L^p(\Omega_i)} \|g_i\|_{L^q(\Omega_i)} \\
&\leq \sum_{i \geq 0} C 2^{i(\gamma-1)} \|M_{res, \Omega_i, \sigma}^{\#} f\|_{L^p(\Omega_i)} \|g_i\|_{L^q(\Omega_i)} \\
&\leq \sum_{i \geq 0} C 2^{i(\gamma-1)} \|M_{res, \Omega, \sigma}^{\#} f\|_{L^p(\Omega_i)} \|g_i\|_{L^q(\Omega_i)}.
\end{aligned}$$

Finally, using the Hölder inequality in the sum, the fact that the weight is locally quasi-constant, and estimation (3.14), we have

$$\begin{aligned}
\int_{\Omega} f(x)g(x) \, dx &\leq C \left(\sum_{i \geq 0} \int_{\Omega_i} \left(d_0(x)^{1-\gamma} M_{res, \Omega, \sigma}^{\#} f(x) \right)^p \, dx \right)^{1/p} \left(\sum_{i \geq 0} \|g_i\|_{L^q(\Omega_i)}^q \right)^{1/q} \\
&\leq C \left(\sum_{i \geq 0} \int_{\Omega_i} \left(d_0(x)^{1-\gamma} M_{res, \Omega, \sigma}^{\#} f(x) \right)^p \, dx \right)^{1/p} \|g\|_{L^q(\Omega)}^q \\
&\leq C \|d_0^{1-\gamma} M_{res, \Omega, \sigma}^{\#} f\|_{L^p(\Omega)}.
\end{aligned}$$

□

Theorem 3.4.5. Let $\Omega \subset \mathbb{R}^2$ be the domain introduced in (3.9) and $1 \leq p < \infty$. Given $\sigma \geq 1$ and $\alpha > -(\gamma + 1)/p$, there exists a constant $C > 0$ such that

$$\|d_0^{\alpha} f\|_{L_0^p(\Omega)} \leq C \|d_0^{1-\gamma+\alpha} M_{res, \Omega, \sigma}^{\#} f\|_{L^p(\Omega)}, \quad (3.22)$$

for any $f \in L_0^p(\Omega, d_0^{p\alpha})$, where $d_0(x) = \sqrt{x_1^2 + x_2^2}$.

Proof. This result follows the same idea used in Theorem 3.4.4 with the locally quasi-constant power weights discussed in Example 3.3.2. Notice that the condition on α is due to the fact that we need the weights to be locally quasi-constant. □

List of Symbols

M	The Hardy-Littlewood maximal
$M_{res,\Omega,\sigma}$	The restricted maximal function
$M_{res,\Omega,\sigma}^\sharp$	The restricted sharp maximal on $\Omega \subset \mathbb{R}^n$ and $\sigma \geq 1$
Q	A cube in $\mathbb{R}^n$
Ω	bounded domain in $\mathbb{R}^n$
$l(Q)$	The side length of a cube
$B(x, r)$	An open ball centered at x with radius r
$\overline{B}(x, r)$	A closed ball centered at x with radius r
f_A	The average value of a function f over a set $A \subset \mathbb{R}^n$
$ A $	Lebesgue measure of a set $A \subset \mathbb{R}^n$
∂U	The boundary of a set $U \subset \mathbb{R}^n$
$L^p(\Omega)$	The space of functions such that $(\int_\Omega f ^p)^{\frac{1}{p}} < \infty$
$L_0^p(\Omega)$	The space of functions in $L^p(\Omega)$ with mean value zero
$L_{loc}^p(\Omega)$	The space of all functions such that $(\int_K f ^p)^{\frac{1}{p}} < \infty$ for every compact $K \in \Omega$
$W^{1,p}(\Omega)$	The first order Sobolev space

Definitions

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